



Navigating IL-6: From molecular mechanisms to therapeutic breakthroughs

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ABSTRACT

This concise review navigates the intricate realm of Interleukin-6 (IL-6), an important member of the cytokine family. Beginning with an introduction to cytokines, this narrative review unfolds with the historical journey of IL-6, illuminating its evolving significance. A crucial section unravels the three distinct signaling modes employed by IL-6, providing a foundational understanding of its versatile interactions within cellular landscapes. Moving deeper, the review meticulously dissects IL-6's signaling mechanisms, unraveling the complexities of its pleiotropic effects in both physiological responses and pathological conditions. A significant focus is dedicated to the essential role IL-6 plays in inflammatory diseases, offering insights into its associations and implications for various health conditions. The review also takes a therapeutic turn by exploring the emergence of anti-IL-6 monoclonal inhibitors, marking a profound stride in treatment modalities. Diving into the molecular realm, the review explores small molecules as agents for IL-6 inhibition, providing a nuanced perspective on diverse intervention strategies. As the review embarks on the final chapters, it contemplates future aspects, offering glimpses into potential research trajectories and the evolving landscape of IL-6-related studies.

1. Introduction

Cytokines, the multifaceted signaling molecules of the immune system, are fundamental contributors to the intricate tapestry of human biology. Evolving to interpret environmental cues vital for maintaining physiological balance, these secretory proteins extend their influence far beyond the realm of immune homeostasis and the inflammatory response to infections, trauma, or injury. From embryonic development to cognitive functions, tissue integrity, and the aging process, cytokines exercise their diverse functions, often displaying pleiotropic and

overlapping properties that defy conventional categorization [1,2]. As humoral factors orchestrating intercellular communication, cytokines are essential for sustaining homeostasis. Their roles have been scrutinized through in vitro cell stimulation experiments, probing protein interactions, and in vivo exploration using genetically engineered mice and disease models. This in-depth investigation has unearthed the central role of the cytokines which play in the pathophysiology of various diseases, propelling the development of pharmacotherapies targeting specific cytokines through biologic drugs [3,4]. This profound understanding has transformed the outcome of chronic inflammatory diseases,

Abbreviations: IL-6, Interleukin-6; TNF- α , Tumor Necrosis Factor-alpha; LPS, Lipopolysaccharide; iNOS, Inducible Nitric Oxide Synthase; AP, Andrographolide; XYP, Xiyanping; COX-2, Cyclooxygenase-2; MDL-A, Madindoline A; BG, Bergenin; IC₅₀, Half-Maximal Inhibitory Concentration; MPO, Myeloperoxidase; HSV-1, Herpes Simplex Virus Type 1; HIV, Human Immunodeficiency Virus; CCK-8, Cell Counting Kit-8; GP130, Glycoprotein 130; MAPK, Mitogen-Activated Protein Kinase; SOCS, Suppressor of Cytokine Signaling; ERK, Extracellular Signal-Regulated Kinase; JNK, c-Jun N-terminal Kinase; NF- κ B, Nuclear Factor-kappa B; ICAM-1, Intercellular Adhesion Molecule-1; MCP-1, Monocyte Chemoattractant Protein-1; PARP, Poly(ADP-ribose) Polymerase; Bcl-2, B-cell lymphoma 2; Bax, Bcl-2-associated X protein; Akt, Protein Kinase B; mTOR, Mammalian Target of Rapamycin; STAT3, Signal Transducer and Activator of Transcription 3; ROS, Reactive Oxygen Species; IL-1 β , Interleukin-1 beta; TLR4, Toll-like Receptor 4; PPAR- γ , Peroxisome Proliferator-Activated Receptor gamma; NFAT, Nuclear Factor of Activated T cells; AMPK, AMP-activated Protein Kinase; I κ B, Inhibitor of kappa B; NLRP3, NOD-like Receptor Protein 3; TGF- β , Transforming Growth Factor-beta; VEGF, Vascular Endothelial Growth Factor; Sgp130Fc, soluble glycoprotein 130Fc Fusion protein.; OSM, Oncostatin M; LIF, Leukaemia inhibitory factor; CNTF, Ciliary neurotrophic factor; CT-1, Cardiotrophin 1; CLCF1, Cardiotrophin-like cytokine factor 1.

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with cytokine inhibitors emerging as transformative agents [5]. Soluble regulators facilitating intercellular dialogue in the realms of immune responses and hematopoiesis, cytokines wield their influence with remarkable functional pleiotropy and redundancy. Though crucial for host defense and tissue homeostasis, aberrant cytokine production can trigger the onset and progression of acute and chronic diseases, thereby inspiring innovative therapeutic strategies using cytokines themselves or precision-targeted biologics [6,7]. In the vast cytokine landscape, interleukins emerge as central players primarily acting upon leukocytes. These soluble mediators, part of a broader cytokine family, include IFNs, chemokines, lymphokines, TGF- β , colony-stimulating factors, and TNF. Their functional diversity, alongside characteristic redundancy, serves as a cornerstone of immune communication [8]. Remarkably compact, cytokines, with their 15–20 kDa size, hold the key to modulate homeostasis and inflammation. Pioneering efforts to decipher individual cytokines and their intricate signaling cascades have paved the way for clinical therapies addressing immune disorders [2,9,10]. Cytokines, while diverse in function, often share structural homology, belonging to the four α -helical class of mediators characterized by common up-up-down-down topology in their four helices. Organized into families based on the structure and the composition of their receptor complexes, cytokines bind to multimeric receptor complexes, frequently featuring subunits common to other cytokine receptor complexes. This intricate web of signaling molecules embodies the essential language of the immune system, constantly evolving and expanding the frontiers of our understanding of human health and disease [11,12].

2. Interleukin-6 and its history

Interleukin-6 (IL-6) emerges as a multifaceted proinflammatory cytokine, playing a essential and multifunctional role in orchestrating immune system dynamics with far-reaching implications for immune regulation [13,14]. An aberrant expression or dysregulation of IL-6 has been closely linked to the onset and perpetuation of chronic inflammatory states, as well as the intricate pathogenesis of multifactorial autoimmune disorders [15–17]. The IL-6 cytokine family encompasses a diverse array of members, including IL-6, IL-11, IL-27, IL-31, oncostatin M (OSM), leukaemia inhibitory factor (LIF), ciliary neurotrophic factor (CNTF), cardiotrophin 1 (CT-1), and cardiotrophin-like cytokine factor 1 (CLCF1). Arguably, among all cytokine families, the IL-6 family exhibits the highest degree of functional pleiotropy and redundancy, eliciting responses relevant to both health and disease. These cytokines play important roles in chronic inflammation, autoimmunity, infectious diseases, and cancer often serving as diagnostic or prognostic indicators of disease activity and responsiveness to therapy. Furthermore, the IL-6 family cytokines have emerged as key targets for clinical intervention, emphasizing their critical roles in various physiological and pathological processes [2,18–21]. The Table 1. serves as a comprehensive guide to the IL-6 cytokine family, offering a concise overview of each member's gene symbol, cellular expression, functions, approved receptors, and roles in physiological and disease processes. It acts as a valuable resource for researchers and readers, providing a quick reference to understand the diverse functions of these cytokines. Notably, it highlights their significance in various biological processes such as immune responses, tissue homeostasis, hematopoiesis, and metabolism. The clear categorization of cytokines based on their expression and involvement in different physiological functions allows for easy interpretation. This table is an essential tool for researchers in immunology, cellular biology, and clinical medicine, facilitating a deeper understanding of the IL-6 cytokine family and its implications in health and disease.

In individuals without underlying health issues, blood IL-6 levels typically range from 1 to 5 pg/ml. During periods of inflammation, these levels can skyrocket by approximately 1,000-fold [11]. Notably, in cases of severe sepsis, IL-6 concentrations have been reported to reach several micrograms per milliliter [22]. In normal, balanced conditions, soluble IL-6 receptor (sIL-6R) levels in the blood fall within a range of 40–75

ng/ml. During inflammatory conditions, they can rise by a factor of 2 to 10-fold [23]. It remains uncertain whether the levels of sIL-6R are directly linked to the severity of diseases, as this aspect has not been comprehensively investigated. However, recent research on patients with inflammatory bowel disease and COVID-19 has produced conflicting findings in this regard [24,25]. In individuals with autoimmune diseases like Rheumatoid Arthritis, the concentrations of IL-6 can significantly escalate, reaching levels as high as 150 ng/ml. This elevation underscores the pivotal role of IL-6 in the pathogenesis of autoimmune conditions, where its dysregulation contributes to chronic inflammation and immune system dysfunction [26]. Interleukin-6 (IL-6) is a 184 amino acid glycosylated protein with a molecular weight ranging from 21 to 28 kDa. It is synthesized and secreted by a wide array of cell types, including monocytes, T-cells, fibroblasts, and endothelial cells [27,28]. IL-6 exerts its diverse functions through binding to the Interleukin-6 receptor (IL-6R), an 80 kDa type I transmembrane protein [29]. This IL-6/IL-6R complex is non-signaling-competent by itself. However, upon binding, it associates with a second receptor subunit, glycoprotein 130 (gp130). This interaction induces dimerization of gp130, which in turn activates several intracellular signaling pathways, such as JAK/STAT, YAP, SHP2/Ras/MAPK. IL-6 plays a crucial role in the transition between acute and chronic inflammation, and its impact on both proinflammatory and anti-inflammatory processes makes it a central player in the regulation of immune responses in humans. Understanding the complex signaling pathways activated by IL-6 is crucial for elucidating its multifaceted roles in various physiological and pathological contexts [1,21,30,31].

3. Journey of IL-6 in 1970s

In the 1970 s, Interleukin-6 (IL-6) found its origins when Kishimoto's group first identified it as a soluble protein generated by T cells. Initially recognized for its role in activating the differentiation of B cells into antibody-producing cells, IL-6 was initially designated as B cell stimulatory factor 2. This important discovery laid the foundation for subsequent research into IL-6, unraveling its multifaceted functions and positioning it as a central player in the intricate network of immune regulation [5,6].

3.1. Journey of IL-6 in 1981-1990

In 1986, the significant milestone of cloning IL-6 cDNA marked a pivotal moment in understanding its molecular structure. Subsequent years saw crucial advancements: the cloning of IL-6R cDNA in 1987, the identification of sIL-6R in 1989, and the cloning of gp130 cDNA in 1990 [8,10].

3.2. Journey of IL-6 in 1991-2000

The year 1992 brought about transformative developments with the generation of anti-IL-6R (mouse) and the revelation that sIL-6R could be produced through alternative splicing in humans and shedding. In 1993, the advent of anti-IL-6R (tocilizumab, humanized) further expanded therapeutic possibilities. The year 1994 witnessed the creation of an IL-6 knockout mouse and the inception of 'trans-signaling' conceptualization. Advancements continued with the branding of IL-6 trans-signaling in 1997 and the development of IL-6-sIL-6R trans-signaling transgenic mice. By 1998, IL-6's role in murine rheumatoid arthritis (RA) was elucidated. The new millennium ushered in breakthroughs such as the blocking of murine inflammatory bowel disease (IBD) by sgp130Fc in 2000 [10,23,32].

3.3. Journey of IL-6 in 2001-2010

In 2001, significant progress was achieved with the selective inhibition of IL-6 trans-signaling by sgp130Fc. This development showcased

Table 1

IL-6 cytokine family, offering a concise overview of each member's gene symbol, cellular expression, functions, approved receptors, and roles in physiological and disease processes.

Approved name	Gene symbol	Cellular expression	Function	Approved Receptor and gene symbol	Cellular expression	Physiological and disease processes. High/Medium/Low	IL-6 family cytokines as regulators of metabolic processes. High/Medium/Low
IL-6	IL-6, BSF2, HGF, HSF	Macrophage, DC, lymphocyte, epithelial cell, osteoclast, hepatocyte	Acute responses, angiogenesis, osteoclastogenesis, differentiation of Th17 subset and B cell, glucose metabolism	IL-6 receptor (IL-6R) and CD126 IL-6 signal transducer (IL-6ST) and GP130, CD130, sGP130	Macrophage, monocyte, DC, hepatocyte, adipocyte Ubiquitous expression	High: Innate immunity, Adaptive immunity, Oncogenesis, Cancer, Autoimmunity, Inflammation. Medium: Tissue homeostasis, Haematopoiesis, Pain. Low: Mental well-being, Neurotrophic factor.	High: Bone metabolism, Energy metabolism, Glucose metabolism, Lipid metabolism, Iron transport, Thermogenesis
IL-11	IL-11, AGIF	Stromal cell line, fibroblast, chondrocytes, various cancers	Hematopoiesis, adipogenesis, neuronal differentiation, bone metabolism, cell proliferation, invasiveness	IL-11 receptor subunit α (IL11RA)	Various cancers	High: Innate immunity, Tissue homeostasis, Oncogenesis, Cancer, Autoimmunity, Inflammation. Medium: Adaptive immunity, Haematopoiesis, Pain. Low: Mental well-being, Neurotrophic factor	High: Bone metabolism. Medium: Energy metabolism. Low: Iron transport.
Oncostatin M (OSM)	MGC20461	Monocyte, macrophage, neutrophil, T cell	Hematopoiesis, bone turnover, lipid metabolism, liver regeneration	Oncostatin M receptor (OSMR) and OSMRB, OSMR β	Nonhematopoietic cell; hepatocyte, epithelial cell, endothelial cell, stromal cell, fibroblast cell	High: Innate immunity, Tissue homeostasis, Haematopoiesis, Autoimmunity, Inflammation. Medium: Cancer, Adaptive immunity, Oncogenesis, Pain. Low: Neurotrophic factor	High: Bone metabolism. Medium: Energy metabolism, Glucose metabolism, Lipid metabolism, Iron transport, Thermogenesis.
Leukemia inhibitory factor (LIF)	CDF, DIA, HILDA	T cell, activated monocyte, fibroblast, endothelial cell	Bone remodeling, neural regeneration	LIF receptor subunit alpha (LIFR) and CD118	skeletal muscle cell, cardiomyocyte	High: Tissue homeostasis, Oncogenesis, Neurotrophic factor Medium: Innate immunity, Adaptive immunity, Haematopoiesis, Pain, Cancer, Inflammation. Low: Autoimmunity	High: Bone metabolism. Medium: Energy metabolism, Glucose metabolism, Lipid metabolism, Iron transport. Low: Thermogenesis.
Cardiotrophin 1 (CTF1)	CT-1, CT1	Cardiac myocyte	Apoptosis			High: Tissue homeostasis Low: Adaptive immunity, Haematopoiesis, Innate immunity, Autoimmunity, Inflammation.	High: Energy metabolism. Medium: Glucose metabolism, Lipid metabolism. Low: Bone metabolism, Thermogenesis.
Ciliary neurotrophic factor (CNTF)	HCNTF	Osteoblast, osteocyte, osteoclast, chondrocyte	Bone metabolism, glucose metabolism	Ciliary neurotrophic factor receptor (CNTFR)	Skeletal muscle cell	High: Tissue homeostasis, Haematopoiesis, Neurotrophic factor Medium:	High: Bone metabolism. Medium: Energy metabolism,

(continued on next page)

Table 1 (continued)

Approved name	Gene symbol	Cellular expression	Function	Approved Receptor and gene symbol	Cellular expression	Physiological and disease processes. High/Medium/Low	IL-6 family cytokines as regulators of metabolic processes. High/Medium/Low
Cardiotrophin-like cytokine factor 1 (CLCF1)	NNT1, BSF3, CLC, NR6, CISS2, BSF-3, NNT-1	Activated Jurkat human T cell lymphoma cells	Development of nervous system			Innate immunity, Adaptive immunity, Pain, Inflammation. Low: Mental well-being, Autoimmunity High: Tissue homeostasis, Medium: Innate immunity, Adaptive immunity, Neurotrophic factor Low: Haematopoiesis, Pain, Inflammation, Autoimmunity.	Glucose metabolism, Lipid metabolism, Low: Thermogenesis. Medium: Bone metabolism, Energy metabolism.
IL-27	IL-27, p28, IL27p28, IL-27A, IL27A, MGC71873	Antigen-presenting cell	Differentiation of T cell subset	IL-27 receptor subunit α (IL27RA) and WSX-1, CCR, CRL1, WSX1, zcytor1, IL-27R	Macrophage, DC, T cell, B cell	High: Innate immunity, Adaptive immunity, Autoimmunity, Inflammation. Medium: Tissue homeostasis, Haematopoiesis, Oncogenesis, Cancer, Pain, Low: Mental well-being, Neurotrophic factor High: Innate immunity, Tissue homeostasis. Medium: Autoimmunity, Inflammation, Adaptive immunity. Low: Oncogenesis, Cancer, Neurotrophic factor.	High: Glucose metabolism Medium: Lipid metabolism Low: Bone metabolism, Iron transport,
IL-31	IL-31			IL-31 receptor subunit-α (IL-31Rα)			

a refined understanding of the intricate signaling mechanisms associated with IL-6, laying the groundwork for more targeted therapeutic interventions. These findings in 2001 underscored the ongoing efforts to elucidate the specific pathways and potential clinical applications related to IL-6 modulation. In 2003, sgp130Fc was revealed to block murine arthritis, marking a significant advancement in understanding its therapeutic potential. The following year, in 2004, Tocilizumab was evaluated in a trial for Crohn’s disease, and sgp130Fc demonstrated effectiveness in blocking murine colon cancer. These years underscored the expanding spectrum of IL-6-related research and therapeutic applications during that period. Notably, sgp130Fc demonstrated improvements in perioperative cognitive disorder in the same year. In subsequent years, the landscape of IL-6 research evolved significantly: Tocilizumab gained approval for castleman disease in 2006. In 2007, Tocilizumab entered trials for rheumatoid arthritis, broadening its clinical applications. In 2008, essential developments included the creation of transgenic mice for systemic sgp130Fc and the introduction of affinity-optimized sgp130Fc, showcasing enhanced efficacy in blocking acute local inflammation. Tocilizumab gained approval for rheumatoid arthritis in 2009. The timeline of innovations persisted with Tocilizumab gaining approval for rheumatoid arthritis in the EU in 2009 and in the USA in 2010 [8,10,23,34].

3.4. Journey of IL-6 in 2011-2022

In 2011, sgp130Fc was shown to block sepsis, and its preventive effects on pancreatic cancer were identified. In 2012, Olamkicept (sgp130Fc) entered clinical phase I for IBD, and its effectiveness in blocking arteriosclerosis was demonstrated. The year 2013 brought insights into sgp130Fc, showing no blockade of bacterial infection, but exhibiting potential in blocking pancreatitis-associated lung failure and lupus erythematosus. In 2014, transgenic mice for brain sgp130Fc were developed, followed by the blocking of nephrotoxic nephritis by sgp130Fc in 2015. In 2016, sgp130Fc was shown to block IL-11 trans-signaling and prevent lung emphysema and lung cancer. The trajectory of IL-6 research remained dynamic, with sgp130Fc blocking murine liver cancer in 2017, highlighting the regulatory role of CNS trans-signaling in body weight. In 2018, novel findings emerged, revealing the role of endogenous sgp130 as a buffer system for IL-6–sIL-6R. Additionally, sgp130Fc demonstrated its ability to enhance bone healing, providing a new dimension to its therapeutic capabilities. In 2019, sIL-6R trans-signaling mice were introduced. These advancements further underscored the multifaceted nature of IL-6-related research, positioning it at the forefront of diverse medical applications. The year 2021 witnessed the emergence of exclusive IL-6 trans-signaling

inhibitors, sgp130^{FlyR} and cs130. Tocilizumab was approved for COVID in 2021, and Olamkicept (sgp130Fc) progressed to clinical phase IIa for IBD [5,6,8,10,23,32–34].

4. Three distinct signaling modes of Interleukin-6

IL-6 orchestrates diverse cellular responses through three distinct signaling modes, each contributing to the intricate biological activities of this multifunctional cytokine. In the classic signaling pathway, IL-6 first binds to its membrane-bound receptor, IL-6R, and subsequently forms a complex with gp130, a ubiquitous signal-transducing protein expressed in most cell types [35]. This interaction, termed classic signaling, is central to antibacterial and resolution activities. However, during inflammatory states, such as those induced by infection, IL-6R undergoes cleavage from the cell surface by the metalloproteases ADAM10/ADAM17. This cleavage results in the generation of a soluble form of IL-6R (sIL-6R), particularly through alternative splicing of mRNA in humans [18,29,36]. The IL-6–sIL-6R complex can then bind to gp130, enabling trans-signaling. This mode triggers rapid proinflammatory responses in various cell types, including endothelial cells, epithelial cells, and smooth muscle cells. A third signaling mode, known as trans-presentation, has been recently identified in antigen-specific interactions between dendritic cells and T cells. In this scenario, the IL-6–mIL-6R complex on dendritic cells signals through gp130 on T cells, initiating pathogenic Th17 responses [33]. Broadly speaking, classic signaling primarily engages in antibacterial and resolution activities, whereas trans-signaling and trans-presentation predominantly mediate proinflammatory responses. These three signaling modes collectively govern the diverse biological activities of IL-6, illustrating its nuanced role in maintaining immune homeostasis and responding dynamically to various physiological and pathological conditions [37–39]. In classic IL-6R signaling, activities are restricted to cells expressing both IL-6R α and gp130. In trans-signaling, IL-6 binds to sIL-6R, creating a complex that can activate gp130 on cell types not typically responsive to IL-6 [40–42]. This process broadens IL-6's target cell repertoire and is considered the primary mechanism for IL-6 involvement in various chronic diseases and cancers. By contrast, IL-6 trans-signaling denotes a process involving IL-6 binding to a soluble form of IL-6R α (sIL-6R), thereby maintaining the circulating half-life of IL-6 and significantly increasing its bioavailability. In this mode, IL-6 can exert its effects on a broader range of cell types that may not express the membrane-bound IL-6R. This broadens IL-6's influence and is considered a key mechanism contributing to its involvement in various chronic diseases and cancers. The dynamic interplay between these signaling modes highlights the versatility and complexity of IL-6 in orchestrating physiological responses in health and disease. Similar trans-signaling mechanisms have been observed for IL-11 and CNTF, while the identification of soluble variants of gp130 (sgp130) further emphasizes the biological importance of these alternative signaling mechanisms. These interactions collectively highlight the versatility and complexity of IL-6 signaling pathways, contributing to its crucial roles in health and disease [30,43]. IL-6, a multifaceted cytokine, engages in a complex network of signaling pathways that profoundly influence diverse physiological processes. Through classical cytokine receptor signaling, IL-6 contributes to acute-phase responses, humoral and T cell-mediated immunity, and cellular proliferation and survival. This traditional mode of signaling extends its reach into processes such as epithelial regeneration and metabolic regulation. In the realm of cytokine trans-signaling, IL-6 assumes a central role in oncogenesis, chronic inflammation, allergies, and immunomodulation [9,21]. Notably, it hinders T cell apoptosis, facilitates leukocyte recruitment, and influences cellular differentiation, thereby modulating immune responses. Moreover, IL-6 participates in tissue remodeling, regeneration, and fever induction, reflecting its broad impact on tissue homeostasis and systemic responses. The unique IL-6 trans-presentation mode adds another layer of complexity, contributing to T cell commitment, dendritic cell activity, and

neuroinflammatory processes. As a master regulator, IL-6 emerges as a key orchestrator in a wide spectrum of physiological and pathological contexts, underscoring its essential role in maintaining homeostasis and responding to dynamic challenges [23,44–46]. When IL-6 engages with either the transmembrane IL-6 receptor (IL-6R) or its soluble counterpart (sIL-6R), it instigates a important signaling cascade that diverges into two distinct pathways: the classic and trans-signaling pathways. This signaling initiation prompts the homodimerization of gp130, a crucial signal-transducing component. What sets gp130 apart is its role as a shared signaling partner among various members of the IL-6 cytokine family, including leukemia inhibitory factor (LIF), oncostatin M (OSM), ciliary neurotrophic factor (CNTF), IL-11, cardiotrophin 1 (CTF1), cardiotrophin-like cytokine (CLC), IL-27, and IL-35 [3,47]. The intriguing aspect lies in the hexameric structure formed during this activation process, comprising two molecules each of IL-6, IL-6R, and gp130. Notably, while each cytokine within the IL-6 family interacts with its unique binding receptor, the shared utilization of gp130 for signal transmission demystifies the functional overlap observed across these diverse cytokines. This molecular mechanism provides a profound insight into the long-standing mystery of why members of the IL-6 cytokine family exhibit functional redundancy [10,48–50]. By unraveling the commonality in their signaling pathways, these understanding sheds light on the intricate interplay within the IL-6 family, offering a clear explanation for their overlapping functions and roles in diverse physiological processes which shows in Fig. 1.

5. Signalling mechanisms of Interleukin-6

The intracellular signaling pathways initiated by the binding of IL-6 to its soluble or membrane-bound receptor (sIL-6R or mIL-6R) and subsequent activation of gp130 are crucial in mediating various cellular responses. Three primary signaling pathways downstream of gp130 have been identified, each involving distinct intracellular proteins and cascades [16,51]. The first pathway revolves around the activation of Janus kinase (JAK) and signal transducer and activator of transcription 3 (STAT3). Upon IL-6–IL-6R binding to gp130, JAK is activated, leading to the phosphorylation of specific tyrosine motifs on the intracellular region of gp130. Phosphorylated STAT3 forms a homodimer, translocates into the nucleus, and acts as a transcription factor, regulating gene expression [21,52]. Depending on the ligand, STAT1 may also be phosphorylated, providing cytokine specificity. STAT1 and STAT3 can form a heterodimer that further modulates gene transcription. The second pathway involves the JAK and SHP-MAPK cascade. Activated JAK induces the phosphorylation of tyrosine 759 on gp130, initiating a cascade that involves SHP2 (Src homology 2 domain-containing phosphatase) and the mitogen-activated protein kinase (MAPK) pathway [53]. This pathway leads to hyperphosphorylation of MAPK via adaptor proteins Grb2 and Gab. The JAK-SHP2 signaling cascade also enhances the phosphatidylinositol 3-kinase (PI3K)–Akt cascade, contributing to the activation of NF- κ B [54]. The third pathway focuses on gp130's role in augmenting YAP–NOTCH signaling through the Src kinase family member Yes [55,56]. This interaction directly binds to the cytoplasmic domain of gp130, particularly implicated in carcinogenesis. These signaling cascades play crucial roles in various pathological conditions associated with several diseases. Negative feedback regulation of IL-6R signaling occurs through the activation of STAT3 and suppressor of cytokine signaling (SOCS) proteins. Augmented gp130 activates STAT3, which, in turn, targets the genes for SOCS1 or SOCS3 [57]. These proteins act as negative regulators, with SOCS1 inhibiting the catalytic activity of JAK and SOCS3 interacting with the phosphorylation site on gp130, crucial for terminating its activity. The SOCS protein family collectively serves as a negative-feedback loop, attenuating IL-6R signaling cascades [30,58–61]. Upon activation of multiple signaling mechanisms, cells undergo a complex orchestration of biological processes that collectively dictate their fate and function. The JAK/STAT signaling pathway plays a important role in governing cellular activities

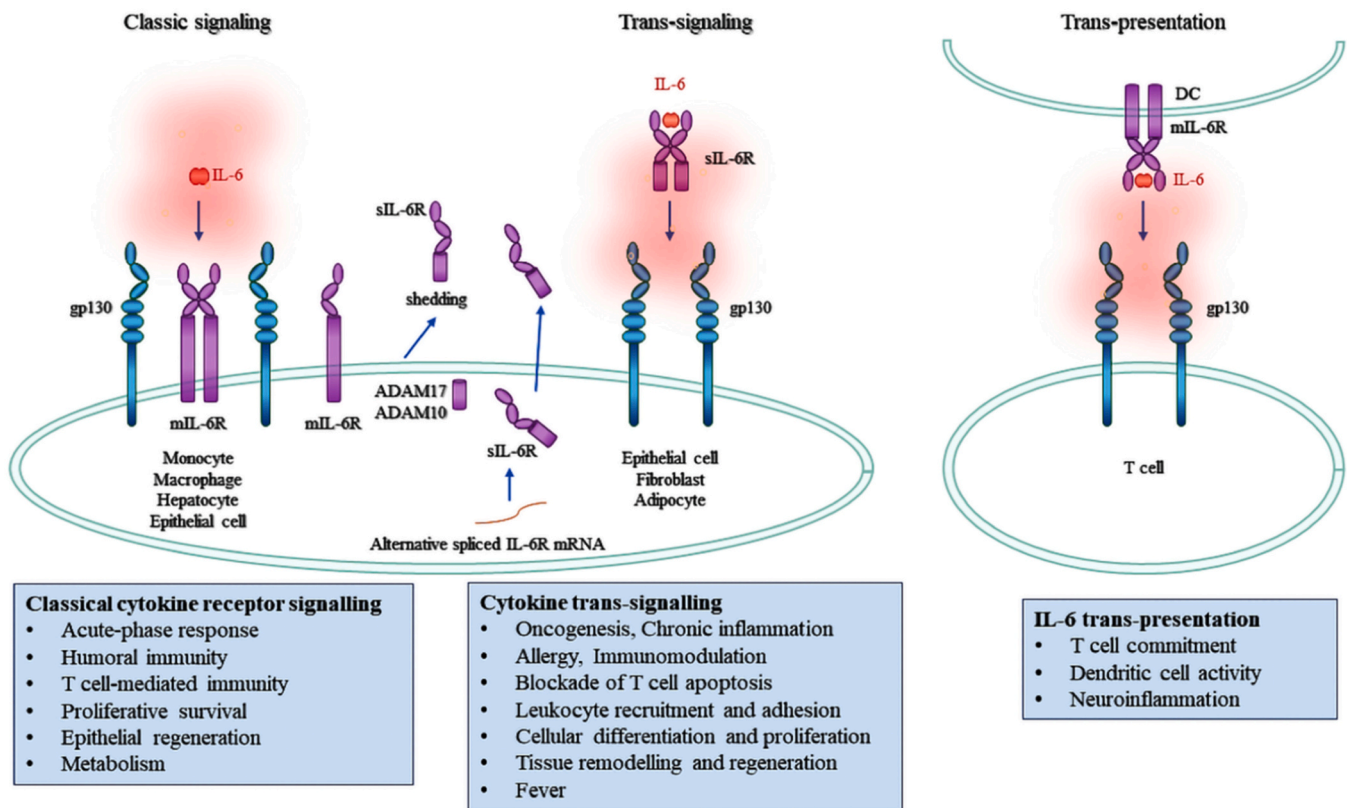


Fig. 1. Three modes of IL-6 interaction: classic signaling via membrane-bound IL-6 receptor (mIL-6R), trans-signaling using soluble IL-6 receptor (sIL-6R), and trans-presentation for antigen presentation. Each mode activates downstream signaling pathways involving gp130.

such as proliferation, survival, differentiation, migration, maturation, self-renewal, immune regulation, angiogenesis, metabolism, and response to oxidative stress. Activation of this pathway initiates a cascade of events influencing diverse cellular functions, from orchestrating immune responses to regulating metabolic processes [23]. Simultaneously, the RAS/RAF/MAPK pathway induces proliferation in cells, contributing to tissue growth and homeostasis. The AKT/PI3K pathway, another critical signaling axis, promotes cellular proliferation, survival, and metabolism while influencing apoptosis and oxidative stress responses. The SRC/NOTCH signaling axis contributes to cellular proliferation and the generation of tissues [6,15]. Collectively, the activation of these signaling mechanisms forms an intricate network that regulates fundamental cellular processes. Proliferation, survival, apoptosis, growth, metabolism, and responses to oxidative stress are finely tuned outcomes resulting from the integrated action of these signaling pathways which is illustrated in Fig. 2 is the canonical IL-6 signaling system, wherein IL-6 activates two primary pathways: the STAT3 and MAPK pathways. These pathways initiate downstream signaling from IL-6R, ultimately resulting in the expression of various genes. Concurrently, this signaling prompts the expression of suppressors of cytokine signaling (SOCS) proteins, specifically SOCS1 and SOCS3, which function to suppress the IL-6 signaling pathway.

IL-6, a pivotal cytokine involved in immunomodulation and inflammation, exerts diverse physiological roles within the human body. It plays an integral part in several crucial activities, contributing to immune responses, tissue development, and homeostasis. By elucidating the following physiological roles of IL-6, we can gain a clearer understanding of its multifaceted functions and its impact on various biological processes. Table 2, explains the pleiotropic effects of IL-6 demonstrate its crucial role in both physiological responses and pathological conditions.

6. The role of IL-6 in inflammatory diseases

IL-6, as a major proinflammatory cytokine, plays a crucial role in the pathogenesis of various inflammatory diseases. Its dysregulated production and signaling contribute to chronic inflammation and tissue damage.

6.1. IL-6 driven rheumatoid arthritis

Rheumatoid arthritis (RA) is a chronic and debilitating inflammatory disorder primarily affecting the synovial membrane, cartilage, and bone, leading to progressive joint damage and functional impairment. The disease is characterized by persistent inflammation within the affected joints, causing pain, swelling, and stiffness. Over time, this inflammatory process can lead to the destruction of cartilage and bone, resulting in joint deformities and reduced mobility [62]. While both rheumatoid arthritis (RA) and osteoarthritis share common symptoms such as joint pain and disability, there are key distinguishing features between the two conditions. One of the characteristic features that sets RA apart from osteoarthritis is the presence of systemic inflammation and elevated levels of IL-6 within the joint fluid. In RA, the immune system is dysregulated, leading to chronic inflammation that extends beyond the affected joints. This systemic inflammation can manifest in various ways throughout the body, affecting multiple organ systems. Elevated levels of IL-6, a pro-inflammatory cytokine, are frequently observed within the joint fluid of RA patients. IL-6 plays a significant role in promoting and sustaining the inflammatory response in RA. In contrast, osteoarthritis is primarily a degenerative joint disease characterized by the gradual breakdown of cartilage and joint structures. While inflammation may occur in osteoarthritis, it is typically localized and less severe compared to RA. IL-6 levels are generally lower in osteoarthritis and do not reach the excessive levels seen in RA. The presence of systemic inflammation and elevated IL-6 levels in the joint fluid of RA patients underscores the

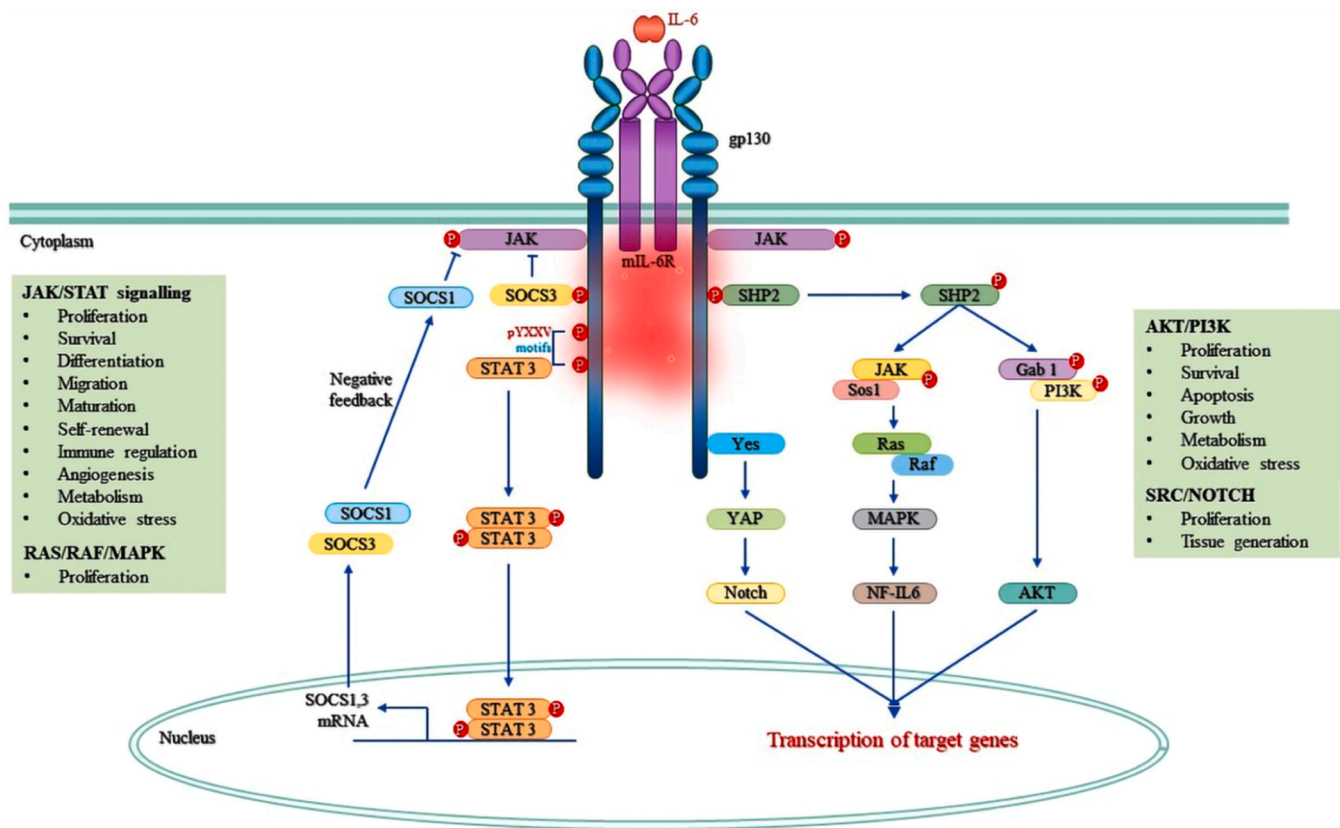


Fig. 2. Schematic representation of the classic IL-6 signaling system. IL-6 activates the STAT3 and MAPK pathways upon binding to its receptor (IL-6R), leading to downstream gene expression. The induced signaling also triggers the expression of suppressors of cytokine signaling (SOCS) proteins, including SOCS1 and SOCS3, which act to dampen the IL-6 signaling cascade.

Table 2
The pleiotropic effects of IL-6 demonstrate its crucial role in both physiological responses and pathological conditions.

Organ & Tissue	Cell types	Effects
Immune system	• CD4 T Cells • CD8 T Cells • B Cells • Plasma blast	• Th17 differentiation↑, • Treg differentiation ↓, • Cytotoxic T cell differentiation ↑, • Antibody production ↑, • Survival ↑
Liver	• Hepatocyte	• CRP, SAA, Hcpidin, fibrinogen↑, • Albumin ↓
Bone	• Synovial fibroblast	• RANKL ↑ means osteoclast differentiation ↑, • VEGF ↑ means angiogenesis and vascular permeability ↑
Haemopoietic system	• Bone marrow cell • Megakaryocyte	• Differentiation ↑ • Platelet release ↑
Skin	• Dermal fibroblast • keratinocyte	• collagen ↑, • myofibroblast differentiation ↑, • proliferation
Kidney	• mesangial cell	• Proliferation ↑
Cardiovascular system	• Vascular endothelial cells • Vascular smooth muscle cell	• Activation ↑ • Proliferation ↑ • Apoptosis ↑

involvement of IL-6 in the pathogenesis of RA [63]. Rheumatoid arthritis (RA) is a complex autoimmune disorder characterized by various immunological abnormalities and systemic inflammation. One prominent aspect of RA is the presence of autoantibodies, including rheumatoid factor and anti-citrullinated protein/peptide antibodies (ACPA). These autoantibodies contribute to the pathogenesis of RA by targeting

self-antigens, such as citrullinated proteins generated through post-translational modification. The multistep process of RA development involves a breakdown of self-tolerance to these citrullinated proteins. This triggers an immune response in T cells and B cells specifically directed against citrulline. The exact mechanisms that lead to the localization of the inflammatory reaction within the joints are still not fully understood. However, once inflammation is established, it initiates and perpetuates synovitis through a positive feedback loop, ultimately leading to systemic inflammation. Patients with RA often exhibit elevated levels of IL-6 in their synovial fluid, directly implicating IL-6 as a crucial factor in the pathogenesis of the disease. Normal levels of IL-6 in human serum are typically less than 4 pg/ml, but in RA and other inflammatory conditions, these levels can rise significantly, ranging from 10 to several hundred pg/ml. Synovial fluid, which is produced within the joints, is known to have higher concentrations of IL-6 compared to serum. The excessive production of IL-6, along with other inflammatory mediators, contributes to the systemic inflammation observed in RA. This chronic inflammatory state leads to various clinical manifestations, including enhanced acute phase protein production, hypoalbuminemia, anemia, and osteoporosis. The systemic effects of RA can result in fatigue, decreased quality of life, and increased mortality rates [64]. IL-6 plays a major role in promoting B cell differentiation into plasma cells, leading to the production of immunoglobulins and subsequent hypergammaglobulinemia and inflammation [65,66]. Additionally, IL-6 directs the differentiation of naive CD4+ T cells. Together with Transforming Growth Factor (TGF)-β, IL-6 induces the differentiation of Th17 cells, which produce IL-17 and TNF-α, while inhibiting the differentiation of TGF-β-induced regulatory T (Treg) cells. This imbalance between Th17 and Treg cells contributes to the dysregulation of immune responses and the development of autoimmune and chronic inflammatory diseases [67,68]. Understanding the role of IL-6 and its relationship

to immunological abnormalities, systemic inflammation, and joint-specific pathologies in RA is critical for developing targeted therapies aimed at modulating IL-6 signaling and managing the disease effectively. By targeting IL-6 and its downstream effects, researchers hope to alleviate inflammation, preserve joint function, and improve the overall well-being of individuals with RA. The Fig. 3 demonstrates the multistep process of RA development, highlighting the role of IL-6 in systemic inflammation, immunological abnormalities, and joint destruction. The elevated levels of IL-6 in the synovial fluid directly contribute to the progression of the disease. The figure provides an overview of the key mechanisms and interactions involving IL-6 in the context of rheumatoid arthritis.

6.2. IL-6 driven cancer

Inflammation plays a significant role in various types of cancers, driving the proliferation and progression of malignant cells. Tumor cells and surrounding cells within the microenvironment secrete inflammatory molecules that promote the rapid growth of cancer cells [69,70]. Within this context, IL-6 has emerged as a key player, with elevated levels of IL-6 associated with continuous tumor growth and resistance to therapy [71,72]. Patients with higher levels of IL-6 in their bloodstream tend to exhibit poorer responses to treatment compared to those with lower levels [73,74]. Moreover, IL-6 is frequently overexpressed in different cancer types, highlighting its strong connection to cancer development [75]. For instance, higher levels of IL-6 in the serum have been linked to an increased risk of liver cancer in males [76]. The activation of the IL-6/STAT-3 signaling pathway is a pivotal event in cancer, governing multiple aspects of the disease [77–79]. Cancer cells, particularly metastatic cells, evade powerful regulatory mechanisms that normally suppress excessive cell proliferation, including the actions of tumor suppressor genes such as P53 and RB. P53 and Retinoblastoma protein (RB) act as checkpoints, deciding whether a cell should

proliferate or undergo senescence and programmed cell death [80]. In cancer, the hypermethylation of CpG islands in the promoter region of tumor suppressor genes enables cells to bypass these regulatory checkpoints during cell cycle progression [81]. IL-6 contributes to this process by stimulating the activity of DNA methyltransferase (DNMT-1), which leads to the increased methylation of CpG islands in the promoter region of P53 genes. IL-6 achieves this by upregulating the expression of DNMT-1 and facilitating its translocation into the nucleus through the phosphorylation of DNMT-1 nuclear localization signal via the PI3K/AKT signaling pathway [82]. Additionally, IL-6 promotes the phosphorylation and inactivation of RB, rendering it unable to bind to E2F and activating cyclin-dependent kinases (CDKs) that drive cell cycle progression into the S phase [83]. Under normal conditions, RB acts as a checkpoint at the transition from G1 to S phase, inducing growth arrest in the G1 phase when active or hypo/dephosphorylated. Understanding the intricate mechanisms by which IL-6 influences cancer development and progression provides valuable insights for the development of targeted therapies. By modulating IL-6 signaling pathways, researchers aim to improve patient outcomes by effectively inhibiting tumor growth and enhancing treatment responses. These findings shed light on the role of IL-6 in tumor growth, its impact on tumor suppressor genes, and the evasion of regulatory mechanisms, paving the way for innovative strategies in cancer treatment.

Fig. 4. visually depicts the impact of IL-6 overexpression and downregulation on cancer growth, highlighting the potential therapeutic implications of targeting IL-6 in cancer treatment. On one side of the figure, it depicts the scenario of IL-6 overexpression. In this case, elevated levels of IL-6 in the tumor microenvironment contribute to tumor progression and growth. IL-6 activates the IL-6/STAT-3 signaling pathway, which regulates multiple aspects of cancer development, including cell proliferation, survival, angiogenesis, and metastasis. The overactive IL-6/STAT-3 signaling promotes the uninterrupted proliferation of cancer cells, evading the normal regulatory mechanisms that

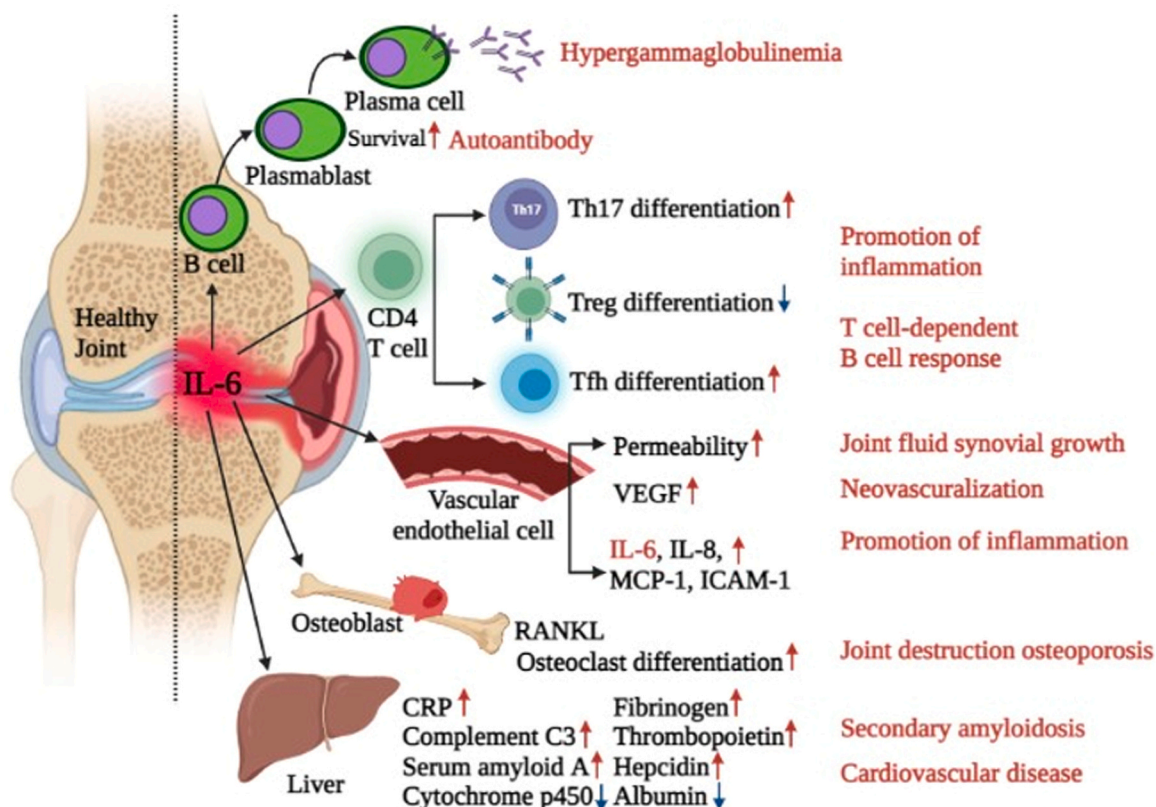


Fig. 3. Illustration depicting the involvement of IL-6 in the pathogenesis of rheumatoid arthritis.

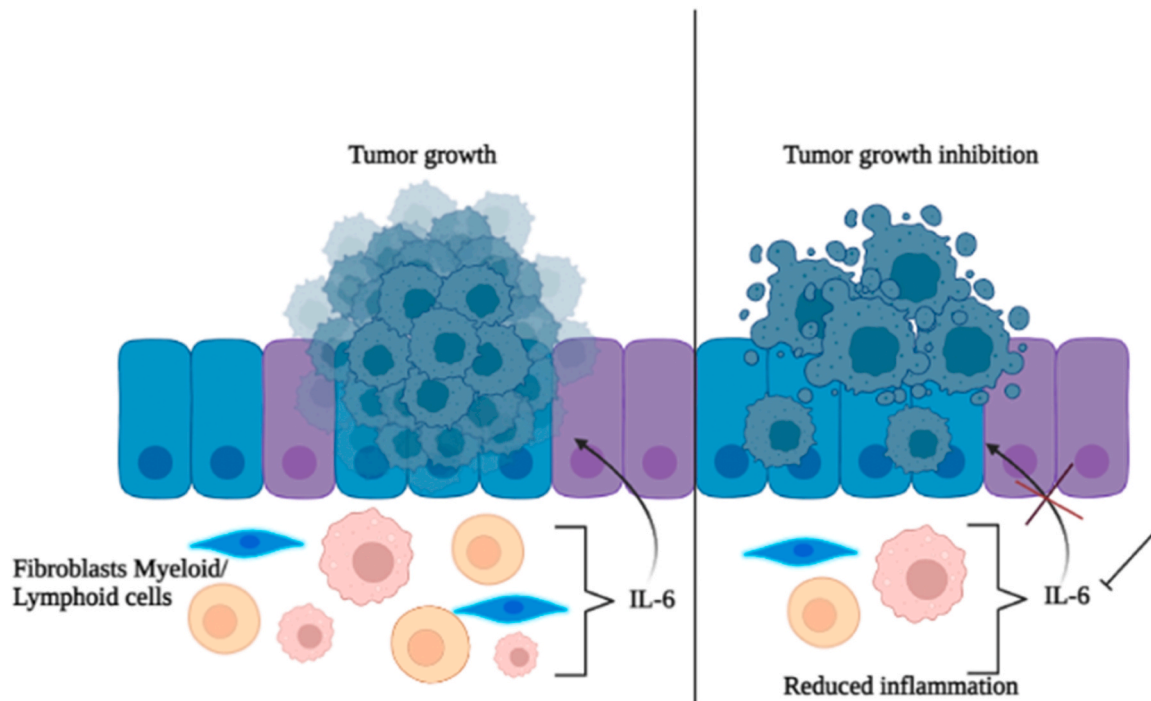


Fig. 4. Role of IL-6 in Cancer Growth.

control cell growth. This leads to increased tumor growth and aggressiveness. On the other side of the figure, it represents the downregulation of IL-6. When IL-6 levels are reduced or blocked, it has a significant impact on inhibiting cancer growth. Downregulation of IL-6 can be achieved through various approaches such as targeted therapies or immunotherapies. Decreased IL-6 levels result in the suppression of the IL-6/STAT-3 signaling pathway, leading to a reduction in cancer cell proliferation, enhanced cell death (apoptosis), and inhibition of tumor angiogenesis and metastasis. These effects collectively contribute to the inhibition of cancer growth and improved treatment outcomes.

6.3. IL-6 driven cytokine storm in COVID-19

The 'cytokine storm' observed in many COVID-19 patients involves the dysregulation and excessive production of various cytokines, including IL-6, IL-1, IL-2, IL-10, TNF- α , and IFN- γ . Among these cytokines, IL-6 has emerged as a significant contributor to the cytokine storm, with increased levels being associated with symptoms such as dyspnea, acute respiratory distress syndrome (ARDS), and poor clinical outcomes [17]. While IL-6 is primarily known for its role as an inflammatory mediator in both innate and adaptive immune responses, it is important to note that it can also exhibit anti-inflammatory and protective properties in certain clinical situations. The complex interplay between IL-6 and other cytokines in the context of COVID-19 highlights the need for a better understanding of the precise mechanisms by which IL-6 influences disease progression and the potential therapeutic implications it holds [49,84]. This cascade of signals has diverse effects on both the acquired immune system (B and T cells) and the innate immune system (neutrophils, macrophages, and natural killer cells), potentially contributing to cytokine release syndrome (CRS). In trans signaling, high levels of circulating IL-6 bind to the soluble form of IL-6 receptor (sIL-6R) and form a complex with a gp130 dimer on various cell types, including endothelial cells. This IL-6-sIL-6R-gp130 complex activates signaling pathways, such as JAK-STAT3, in cells that do not express mIL-6R. This trans signaling further amplifies the "cytokine storm" by inducing the secretion of vascular endothelial growth factor (VEGF), monocyte chemoattractant protein-1 (MCP-1), IL-8, additional IL-6, and

reducing E-cadherin expression on endothelial cells. The secretion of VEGF and reduction of E-cadherin expression contribute to increased vascular permeability and leakage, which play a role in the pathophysiology of hypotension and pulmonary dysfunction, such as acute respiratory distress syndrome (ARDS). These pro-inflammatory effects of IL-6 signaling highlight its involvement in the development and progression of the cytokine storm, which can have detrimental effects on various organ systems during severe inflammatory conditions such as COVID-19. Understanding the mechanisms of IL-6 signaling and its impact on immune and endothelial cells is crucial for developing targeted therapeutic approaches to modulate the cytokine storm and mitigate the associated complications [85]. One particular cytokine that has been consistently found in elevated levels in the bodies of individuals with the disease is IL-6 [86]. Notably, in addition to the elevated levels of IL-6, a crucial observation was the significantly reduced levels of cytokine signaling-3 in these patients. Cytokine signaling-3 plays a vital role in activating the negative feedback mechanism of IL-6, which helps regulate its inflammatory effects. The imbalance between IL-6 and cytokine signaling-3 creates a dysregulated inflammatory response, leading to the exacerbation of symptoms and worsening clinical outcomes. This finding highlights the intricate interplay between IL-6 and its regulatory mechanisms in the context of COVID-19 [87]. Furthermore, an intriguing study unveiled that the heightened levels of IL-6 could potentially serve as a predictive factor for the transition from a mild COVID-19 infection to a more severe and critical condition. The findings suggest that monitoring IL-6 levels in patients could aid in identifying those at risk of disease progression and facilitate early intervention strategies. By closely monitoring IL-6 levels, healthcare professionals can potentially identify individuals who require closer observation and intensive care, allowing for timely interventions and improved patient outcomes. This emphasizes the significance of IL-6 as a potential biomarker for disease severity and highlights its potential utility in guiding clinical management decisions during COVID-19 infection [88]. Several studies have reported the potential efficacy of Tocilizumab, a humanized monoclonal antibody targeting IL-6 receptors, in the treatment of COVID-19. Tocilizumab works by inhibiting the binding of IL-6 to its receptors, thereby reducing the inflammatory

response associated with the cytokine storm. Clinical trials have demonstrated promising results, showing that Tocilizumab can improve clinical outcomes in COVID-19 patients, including reducing the risk of mechanical ventilation and improving survival rates. This therapeutic approach specifically targeting IL-6 highlights the crucial role of this cytokine in the pathogenesis of COVID-19 and offers a potential treatment strategy to mitigate the detrimental effects of excessive inflammation in severe cases [89]. According to a retrospective study conducted in China, the administration of Tocilizumab has demonstrated positive effects in improving fever, reducing C-reactive protein (CRP) levels, and alleviating hypoxemia in COVID-19 patients. Importantly, the study found that Tocilizumab treatment did not lead to any significant adverse effects or complications. These findings suggest that Tocilizumab could be a valuable therapeutic option in managing the inflammatory response associated with COVID-19, particularly in severe cases where patients experience persistent fever, elevated CRP levels, and impaired oxygenation [90]. As per the Italian guidelines, patients who are in the high viral load phase of COVID-19 and exhibit interstitial pneumonia, severe respiratory insufficiency, and elevated levels of IL-6, D-dimer, CRP (C-reactive protein), ferritin, and fibrinogen may be considered for treatment with Tocilizumab. These parameters are used as indicators of severe inflammation and coagulation abnormalities, which are commonly observed in critically ill COVID-19 patients. Tocilizumab, a monoclonal antibody targeting IL-6 receptors, can help mitigate the excessive immune response and cytokine storm associated with severe COVID-19 cases.

Fig. 5. provides an illustration of the progression of events following SARS-CoV-2 infection. The initial target of the virus is the lung alveolar cells, which trigger a robust immune response involving both the innate

and acquired immune systems. This immune response involves the activation and recruitment of various immune cells such as macrophages and monocytes. However, in certain cases, this immune response can become dysregulated, leading to an excessive release of IL-6, a pro-inflammatory cytokine, resulting in a phenomenon known as a cytokine storm. The cytokine storm, characterized by an overwhelming inflammatory response, can have detrimental effects on the body. It can contribute to the development of respiratory acute distress syndrome (ARDS), a severe lung condition characterized by fluid accumulation and impaired oxygen exchange. ARDS, in turn, can lead to respiratory failure and may necessitate mechanical ventilation. Moreover, the cytokine storm and the resulting systemic inflammation can affect multiple organs, potentially leading to multiorgan failure.

6.4. IL-6 driven Alzheimer's Disease (AD)

Alzheimer's Disease (AD) is a neurodegenerative disorder characterized by the accumulation of neurofibrillary tangles (NFTs) and β -amyloid plaques (A β) in the brain [91]. In the context of Alzheimer's Disease (AD), IL-6 production primarily occurs in activated microglia and astrocytes located in different regions of the brain. These cells play a crucial role in the inflammatory response within the brain. Studies have consistently demonstrated elevated levels of IL-6 in the brains of AD patients, as well as in their cerebrospinal fluid and plasma. This suggests that IL-6 is actively involved in the pathological processes associated with AD. The increased presence of IL-6 in AD is believed to contribute to neuroinflammation, which further exacerbates the progression of the disease [92]. The production of IL-6 in the context of Alzheimer's Disease is regulated by specific signaling pathways. In microglia, which are

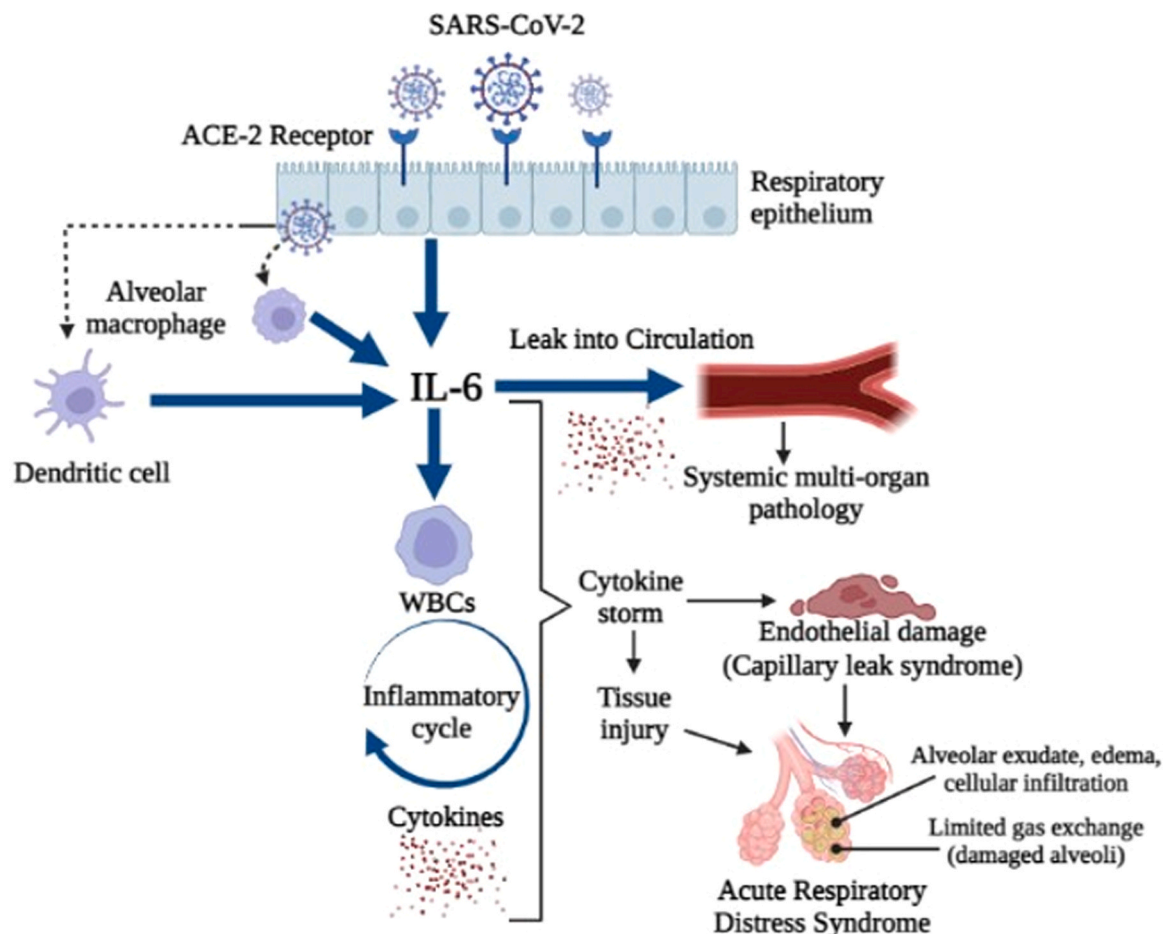


Fig. 5. Role of IL-6 in Cytokine storm in SARS-CoV-2.

immune cells in the brain, IL-6 generation is dependent on the activation of the toll-like receptor 2 (TLR2)-MyD88 signaling pathway. When TLR2 on microglia recognizes certain molecules associated with AD, it triggers a cascade of events that ultimately lead to the production of IL-6. In addition to microglia, endothelial cells in the brain also play a role in IL-6 production. The generation of IL-6 in endothelial cells is regulated by the JNK-AP1 signaling pathway. This pathway is activated in response to various stimuli, including inflammatory molecules and oxidative stress, which are commonly observed in the brains of AD patients. By activating these signaling pathways, TLR2 in microglia and JNK-AP1 in endothelial cells contribute to the production of IL-6 in the brain of individuals with Alzheimer's Disease. The elevated levels of IL-6 further contribute to neuroinflammation and the progression of the disease [93,94]. In addition to microglia and endothelial cells, hippocampal cells also play a role in the production of IL-6 in the context of Alzheimer's Disease. The hippocampus, a region of the brain associated with memory and learning, is known to be affected in AD. Studies have shown that hippocampal cells contribute to the formation of neurofibrillary tangles (NFTs), one of the pathological hallmarks of AD. IL-6 is involved in the process of NFT formation by influencing the phosphorylation of Tau proteins. Tau proteins are normally responsible for stabilizing microtubules in neurons. However, in AD, Tau proteins become abnormally phosphorylated, leading to the formation of tangles. The dysregulation of the cdk5/p35 pathway, which is involved in Tau phosphorylation, can be influenced by IL-6 produced by hippocampal cells. Furthermore, IL-6 activates several signaling pathways, including JAK/STATs, NMDA receptors, and MAPK-p38 protein kinases. These pathways are known to support the hyperphosphorylation of Tau proteins. The activation of these pathways by IL-6 contributes to the pathological process of NFT formation and the progression of Alzheimer's Disease. Overall, the involvement of hippocampal cells in IL-6

production and their impact on Tau phosphorylation provide further insights into the complex mechanisms underlying the development of NFTs and the progression of Alzheimer's Disease [95,96]. In the context of inflammation, an excess of IL-6 can have detrimental effects on cognitive processes such as memory and spatial learning. One mechanism through which IL-6 can exert its negative impact is by activating neuronal NADPH-oxidase. NADPH-oxidase is an enzyme involved in the production of reactive oxygen species (ROS) within neurons. When IL-6 stimulates the activation of NADPH-oxidase, it leads to an increased production of ROS. Excessive ROS production can result in oxidative stress, which can damage neuronal cells and disrupt their normal functioning. The impairment of cognitive processes, including memory and spatial learning, can be attributed to the detrimental effects of IL-6-induced oxidative stress on neurons. The excessive ROS production can lead to oxidative damage to cellular components, including lipids, proteins, and DNA, compromising the integrity and function of neurons. Furthermore, IL-6-mediated inflammation can also disrupt synaptic plasticity, which is crucial for learning and memory processes. IL-6 can interfere with the balance of excitatory and inhibitory neurotransmission, impairing the proper functioning of neuronal circuits involved in cognitive processes. Overall, the activation of neuronal NADPH-oxidase by IL-6 and the subsequent increase in ROS production contribute to impaired cognitive processes observed in conditions of inflammation. Understanding these mechanisms provides insights into the potential therapeutic targets for mitigating the negative effects of IL-6 on cognitive function in diseases like Alzheimer's Disease [97].

Fig. 6 illustrates a cascade of events leading to Alzheimer's Disease (AD). The cascade starts with acute inflammation in microglia, which can be triggered by factors such as aging or stress. This acute inflammation then leads to systemic inflammation, indicating that the inflammatory response extends beyond the brain. As a result of the early

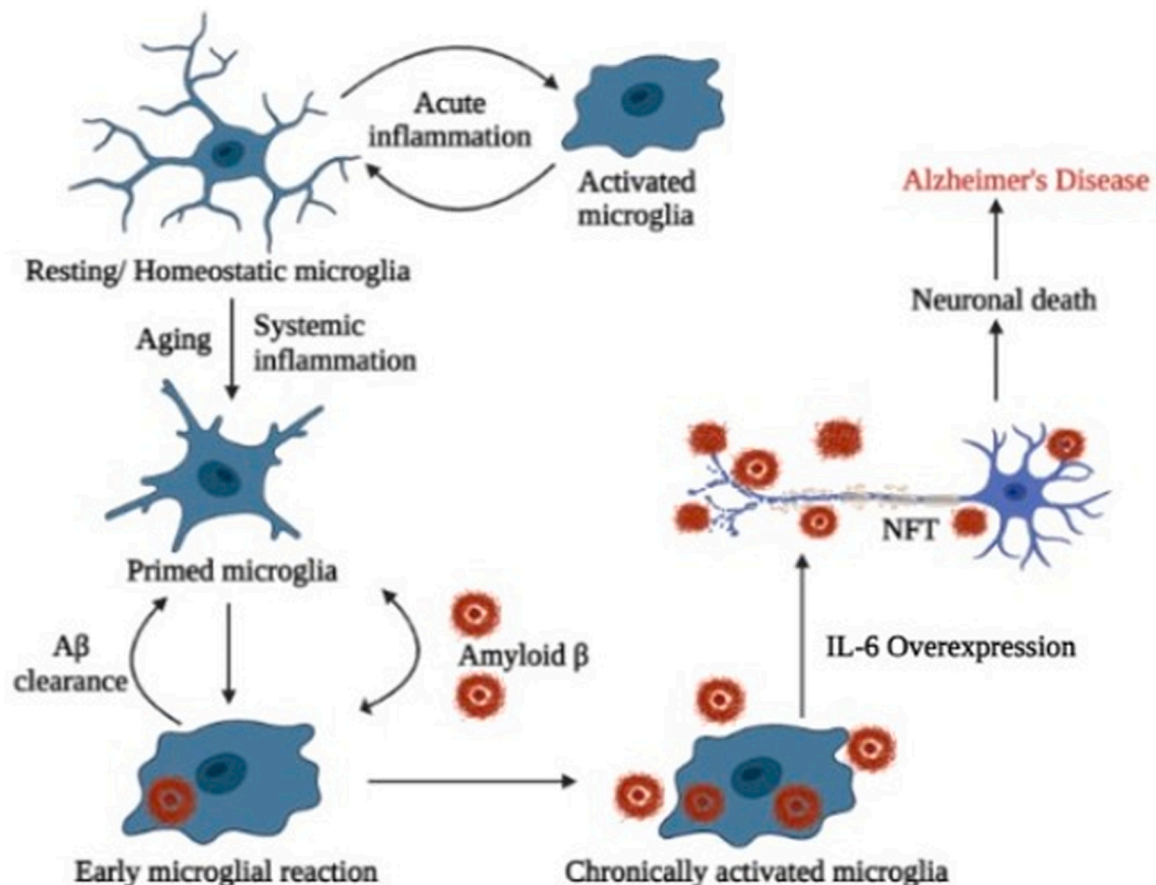


Fig. 6. Role of IL-6 in Alzheimer's Disease (AD).

microglial reaction and chronic activation, the release of amyloid beta ($A\beta$) occurs. $A\beta$ is a key component of the β -amyloid plaques observed in AD. The accumulation of $A\beta$ can contribute to the formation of neurofibrillary tangles (NFTs), which are another hallmark of the disease. The activation of IL-6, possibly driven by the chronic activation of microglia, is suggested to play a role in this cascade. IL-6, as mentioned earlier, is associated with inflammation and can have detrimental effects on neurons. The damage to NFTs in neurons, potentially caused by the chronic activation of IL-6, can lead to neuronal damage and, ultimately, contribute to the development of Alzheimer's Disease.

Inflammatory Bowel Disease (IBD) is a collective term encompassing two major conditions: Crohn's disease and ulcerative colitis. These diseases are characterized by chronic inflammation within the gastrointestinal tract, specifically affecting different segments of the digestive system [98]. Ulcerative colitis primarily affects the colon and is characterized by continuous inflammation of the inner lining (mucosa) and submucosa of the large intestine. On the other hand, Crohn's disease can involve any part of the digestive tract, from the mouth to the anus, and is characterized by patchy inflammation that can affect multiple layers of the intestinal wall, including the mucosa, submucosa, and even the deeper layers (transmural). While Crohn's disease often involves the distal ileum (the last part of the small intestine) and the colon, it can affect other areas as well. Additionally, Crohn's disease can lead to the formation of granulomas, which are clusters of inflamed cells [99]. Indeed, emerging evidence suggests that IL-6 is involved in the development and progression of both Crohn's disease and ulcerative colitis. Studies have shown that patients with active Crohn's disease have significantly higher levels of IL-6 in their bloodstream compared to healthy individuals or those in remission. This indicates that IL-6 may contribute to the ongoing inflammation and symptoms experienced by patients with Crohn's disease. IL-6 is known to be produced by various cells in the gut, including immune cells such as macrophages and T cells, as well as epithelial cells and fibroblasts. Inflammatory stimuli, such as the presence of bacteria or other triggers in the gut, can lead to the activation of these cells and subsequent release of IL-6. Elevated IL-6 levels in Crohn's disease can have several effects. IL-6 promotes the recruitment and activation of immune cells, such as T cells and neutrophils, which can further contribute to the inflammatory response in the gut. It also enhances the production of other pro-inflammatory cytokines and mediators, amplifying the inflammatory cascade. Moreover, IL-6 has been implicated in tissue damage and fibrosis, which are characteristic features of Crohn's disease [100]. In patients with inflammatory bowel disease (IBD), including Crohn's disease and ulcerative colitis, IL-6 plays a significant role in the disease progression. While only a small fraction of lamina propria T cells in patients with inflammatory bowel disease (IBD) express the membrane-bound IL-6 receptors (IL-6R), IL-6 can still exert its effects through an alternative mechanism. IL-6 can bind to soluble IL-6R (sIL-6R), which then forms a complex with gp130, the signal transducing subunit, enabling IL-6 signaling in cells that do not express the membrane-bound IL-6R. The presence of sIL-6R allows IL-6 to activate cells that lack membrane-bound IL-6R, thereby expanding its range of effects. This mechanism is particularly relevant in the context of IBD, where elevated levels of sIL-6R have been observed in the serum of affected individuals. The increased levels of sIL-6R contribute to the formation of the sIL-6R-IL-6 complex, which can circulate in the bloodstream and interact with gp130-expressing cells, including those in the intestinal mucosa. The binding of the sIL-6R-IL-6 complex to gp130 triggers downstream signaling pathways, leading to the activation of various immune cells and the production of pro-inflammatory mediators. This amplifies the inflammatory response within the gut and contributes to the pathogenesis of IBD. The elevated levels of sIL-6R-IL-6 complex in the serum of IBD patients indicate the involvement of IL-6 signaling in the disease. Targeting the IL-6 signaling pathway, including the blockade of IL-6 or sIL-6R, has emerged as a potential therapeutic strategy for IBD. Clinical trials investigating the efficacy of IL-6-targeted therapies, such as anti-IL-6R antibodies, have

shown promising results in reducing inflammation and improving symptoms in IBD patients [101,102]. Fig. 7. illustrates the key mechanisms involved in the pathogenesis of inflammatory bowel disease (IBD). The persistent stimulation of the mucosal immune system by antigens, including the bacterial flora, leads to increased production of IL-6 and soluble IL-6 receptor (sIL-6R) by intestinal macrophages and CD4 + T cells. These cells release IL-6 and sIL-6R into the surrounding environment. The binding of IL-6 to the sIL-6R forms a complex that enhances the expression of STAT3, a transcription factor. Activated STAT3 induces the expression of anti-apoptotic genes, which prevents lamina propria T cells from undergoing apoptosis (programmed cell death). This results in the expansion of T cells within the intestinal tissue. The dysregulated IL-6 signaling pathway and the sustained immune response contribute to the perpetuation of chronic intestinal inflammation in IBD. The excessive production of IL-6, along with the activation of STAT3 and the resistance of T cells to apoptosis, contribute to the inflammatory process in the intestines [103].

6.5. IL-6 Driven psoriasis

Psoriasis is a chronic inflammatory skin disease characterized by the presence of multiple erythematous-squamous plaques. It is considered an immunological disease, with immune system dysregulation playing a significant role in its pathogenesis [104,105]. Pustular psoriasis (PP) is a distinct clinicopathological variant of psoriasis characterized by the presence of pustules on erythematous skin. Unlike other forms of psoriasis, PP is characterized by the predominance of intraepidermal collection of neutrophils, which contributes to the formation of pustules. The presence of pustules is a defining feature that sets pustular psoriasis apart from other types of psoriasis [106,107]. IL-6 and STAT3 polymorphisms play a significant role in the hereditary predisposition to psoriasis. Certain genetic variations in these genes have been associated with an increased susceptibility to develop the condition. These variations can influence the regulation of IL-6 and STAT3 signaling pathways, leading to an imbalance in immune responses and the development of psoriatic lesions [108]. In psoriasis, there is a dysregulation of cytokine production, including elevated levels of IL-6. This cytokine is produced by various immune cells, such as T cells, macrophages, and dendritic cells, as well as keratinocytes in the skin. The increased production of IL-6 contributes to the chronic inflammatory response observed in psoriasis. It promotes the activation and proliferation of immune cells, including T cells, and stimulates the production of other pro-inflammatory cytokines [109]. The elevated serum levels of IL-6 in psoriasis patients reflect the systemic inflammation associated with the disease. Measuring IL-6 levels in the blood can serve as a biomarker to assess the inflammatory activity and severity of psoriasis. Higher IL-6 levels are often correlated with more severe disease manifestations and can guide treatment decisions [110]. In psoriatic plaques, various cell types contribute to the production of IL-6. Human keratinocytes, which are the major cell type in the epidermis, have been identified as a significant source of IL-6 production in psoriasis. The production of IL-6 by keratinocytes can be stimulated by cytokines such as IL-17 or IL-36, which are known to be elevated in psoriatic lesions. These cytokines activate signaling pathways in keratinocytes, leading to the release of IL-6 and other pro-inflammatory mediators [111]. Moreover, recent studies have highlighted the crucial role of IL-6 not only in psoriasis but also in pustular psoriasis, a distinct subtype characterized by the presence of pustules on erythematous skin. IL-6 is implicated in driving the abnormal recruitment of neutrophils into the lesional skin of pustular psoriasis. Neutrophils play a prominent role in the pathogenesis of pustular psoriasis, and their excessive accumulation contributes to the formation of pustules and the inflammatory response in the skin [112]. The dysregulated production of IL-6 in both psoriasis and pustular psoriasis contributes to the chronic inflammation, epidermal hyperproliferation, and disturbed differentiation observed in these conditions. Targeting IL-6 or its signaling pathways has emerged as a potential

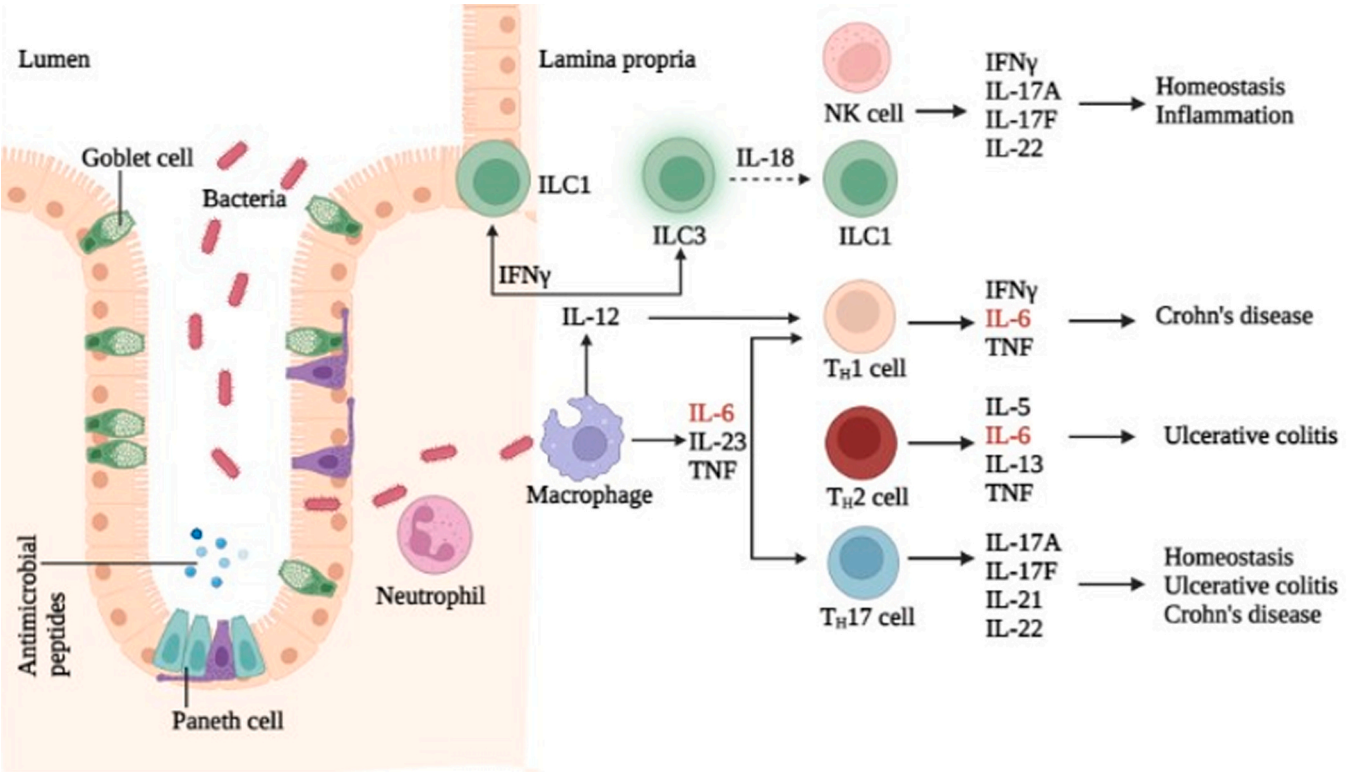


Fig. 7. Role of IL-6 in Inflammatory Bowel Disease (IBD).

therapeutic approach for managing psoriasis and pustular psoriasis, with the aim of reducing inflammation, normalizing keratinocyte function, and restoring skin homeostasis [113].

Fig. 8. illustrates the relationship between environmental stress, keratinocyte activation, IL-6 production, epidermal remodelling, and the development of psoriatic plaques. In this scenario, environmental

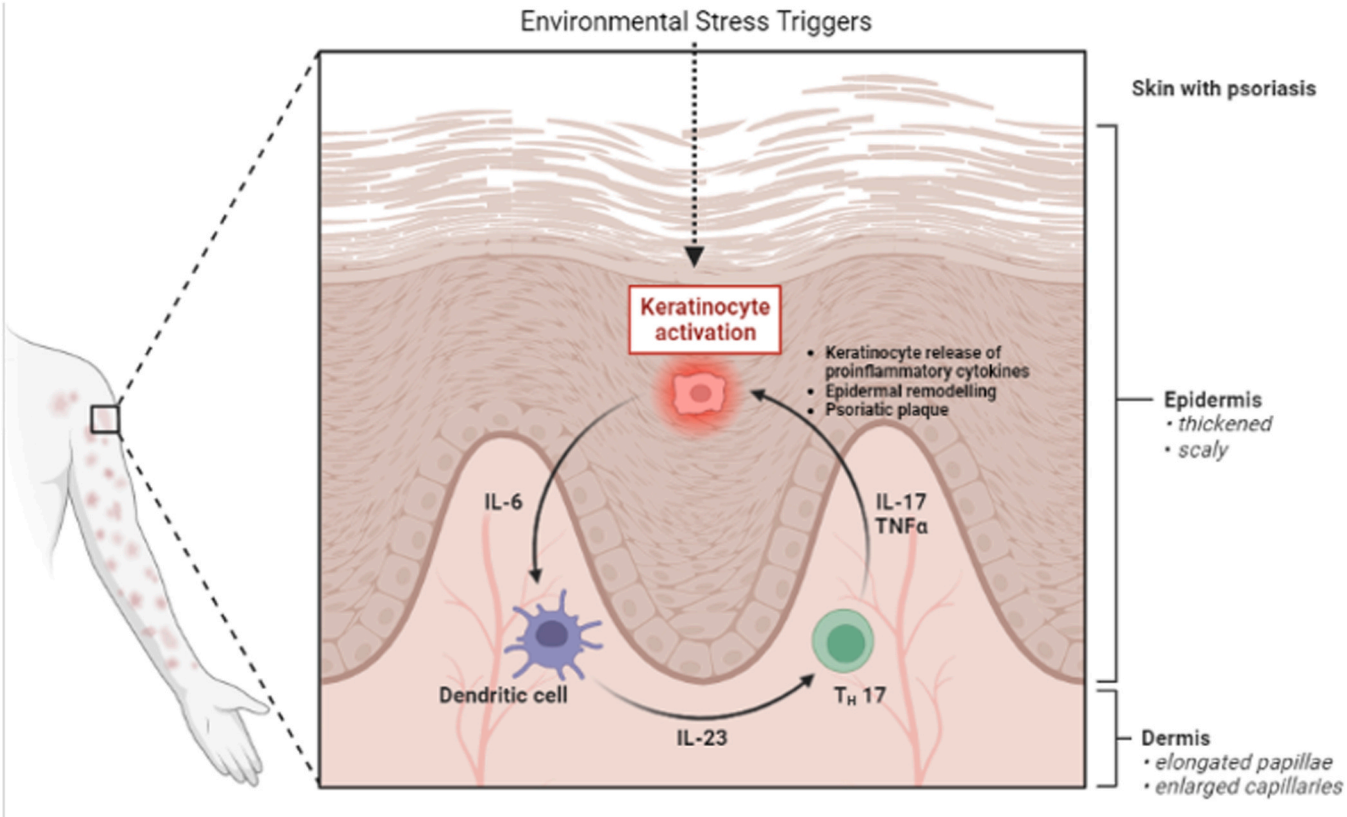


Fig. 8. Role of IL-6 in Psoriasis.

stress factors, such as trauma, infections, or psychological stress, act as triggers on the body. These triggers lead to the activation of keratinocytes, which are the primary cells in the epidermis of the skin. The activated keratinocytes release proinflammatory cytokines, including IL-6, as part of the immune response. IL-6, being a key proinflammatory cytokine, plays a role in promoting inflammation and immune dysregulation in psoriasis. The presence of IL-6 and other inflammatory mediators leads to the remodelling of the epidermis, resulting in abnormal proliferation and differentiation of keratinocytes. This process contributes to the formation of psoriatic plaques, which are characterized by thickened, red, and scaly patches on the skin.

6.6. IL-6 monoclonal inhibitors

IL-6 monoclonal antibodies are a specific class of monoclonal antibodies designed to target and block the activity of interleukin-6 (IL-6), a pro-inflammatory cytokine. IL-6 is involved in various immune responses and plays a critical role in inflammation and autoimmune diseases. These monoclonal antibodies are engineered to specifically bind to IL-6 or its receptors, inhibiting IL-6 signaling and reducing its effects on the immune system. By blocking IL-6, they aim to dampen the excessive immune response and inflammation associated with certain diseases. IL-6 monoclonal antibodies have shown efficacy in the treatment of various conditions where IL-6 plays a significant role, such as rheumatoid arthritis, cytokine release syndrome (CRS) associated with CAR-T cell therapy, and certain inflammatory conditions like Castleman’s disease. These antibodies can be either directed against the soluble form of the IL-6 receptor (sIL-6R), the membrane-bound IL-6 receptor (mIL-6R), or directly against IL-6 itself. By targeting IL-6 or its receptors, these monoclonal antibodies interfere with the IL-6 signaling pathway and help to modulate the immune response. The use of -IL-6 monoclonal antibodies represents a targeted approach to reduce the activity of IL-6 and its downstream effects, providing a potential therapeutic option for patients with IL-6-driven diseases. These antibodies have demonstrated efficacy and safety in clinical trials, leading to their approval for specific indications and ongoing research for potential new applications. Here are some commonly used IL-6 monoclonal antibodies along with their targets, indications, and stages of clinical trials have been summarised in Table 3.

Table 3
IL-6 Monoclonal Antibodies: Targets, Indications, and Clinical Trial Stages.

Name	Target	Indication	Company	Stage Of Clinical Trials	References
Tocilizumab	sIL-6R mIL-6R	Rheumatoid Arthritis, Castleman diseases, Systemic juvenile idiopathic arthritis, Giant cell arteritis, Takayasu arteritis, Adult-onset Still diseases, Cytokine storm in CAR T cell therapy, Severe COVID-19 pneumonia,	Chugai, Roche and Genentech	Approved	[114]
Sarilumab	sIL-6R mIL-6R	COVID-19, Rheumatoid arthritis, Lupus nephritis	Sanofi and Regeneron	III	[115]
Sirukumab	IL-6	COVID-19	Centocor (Janssen)	II	[116]
Clazakizumab	IL-6	Chronic Active Antibody Mediated Rejection in Kidney Transplant, Organ transplant rejection	Bristol Myers Squibb and Alder	III	[117]
Olokizumab	IL-6	Rheumatoid Arthritis, Crohn disease.	Conaris Research Insititute, Ferring Pharmaceuticals and I-Mab Biopharma	III	[118]
Vobarilizumab	IL-6 R	Rheumatoid arthritis and systemic lupus erythematosus	Ablynx	II	[54]
ALX-0061	sIL-6R mIL-6R	Rheumatoid Arthritis		II	[119]

7. Small molecules used for IL-6 inhibition

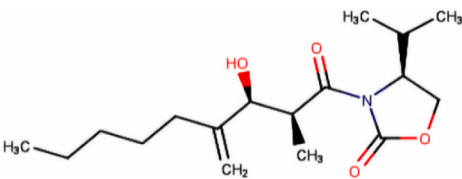
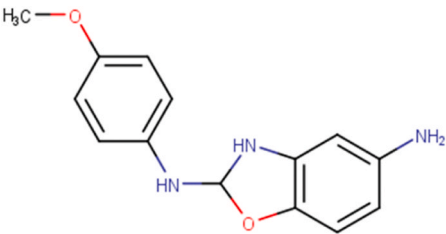
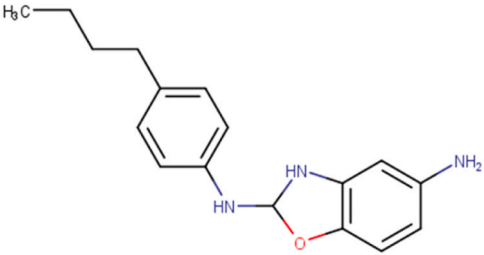
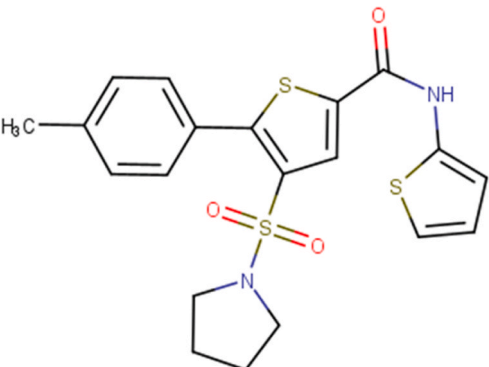
Small molecules used for IL-6 inhibition play a crucial role in the treatment of various diseases, particularly those associated with excessive IL-6 activity. These molecules include both traditional disease-modifying anti-rheumatic drugs (DMARDs) and newer biological DMARDs. Traditional DMARDs, such as methotrexate, sulfasalazine, and leflunomide, have been used in the treatment of rheumatoid arthritis (RA) and other autoimmune conditions. While these drugs provide symptomatic relief, they do not specifically target IL-6 or its signaling pathway. On the other hand, biological DMARDs, including monoclonal antibodies and fusion proteins, have revolutionized the treatment of IL-6-related diseases. These synthetic molecules are designed to specifically inhibit IL-6 or its receptor, thereby suppressing its activity and downstream signaling [120,121]. Some of compounds from the various research literature we identified that these compounds having inhibitory effect of interleukin-6.

7.1. LMT-28

LMT-28 is a small synthetic molecule which is shown in Table 4, Compound (1), that has shown potential as an inhibitor of IL-6-mediated signaling. LMT-28, a novel synthetic compound identified from an in-house chemical library, emerges as a promising small-molecule inhibitor targeting the IL-6 signaling pathway. Through meticulous screening, researchers found that LMT-28 (Oxazolidinone derivative) selectively inhibits IL-6-induced activation of STAT3, showcasing its potential as an antagonist for IL-6-mediated signaling cascades. The compound not only downregulates the phosphorylation of crucial proteins, including gp130 and JAK2, but also significantly hinders IL-6-dependent TF-1 cell proliferation, indicating a robust inhibitory effect on cellular responses triggered by IL-6. Beyond its molecular actions, LMT-28 demonstrates therapeutic promise in vivo, alleviating conditions associated with inflammatory responses such as collagen-induced arthritis and acute pancreatitis in murine models. These findings underscore LMT-28 as a novel synthetic IL-6 inhibitor with potential applications in mitigating inflammatory diseases. Crucially, LMT-28’s mechanism of action involves direct binding to gp130, a key component of the IL-6 signaling pathway. Surface plasmon resonance analysis confirmed the compound’s ability to specifically reduce the binding of IL-6/IL-6Ra complexes to gp130. This direct interaction with gp130 positions LMT-28 as a unique antagonist, disrupting the IL-6

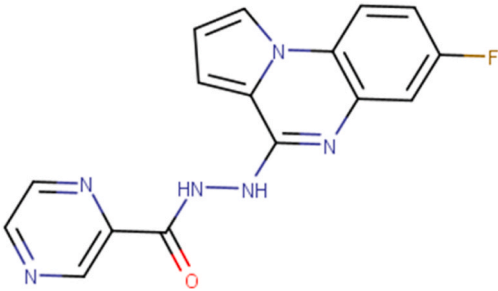
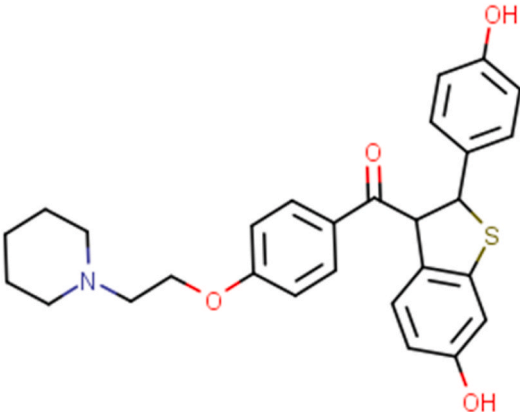
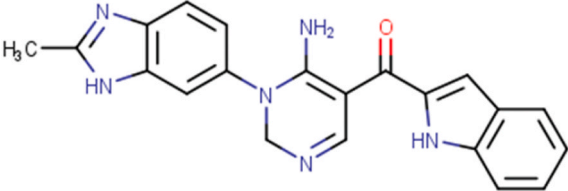
Table 4

Small molecules inhibit IL-6 in different cell lines and their structures.

Compounds	Cell Lines used for IL-6 inhibition.	Structure
Compound (1)	LMT-28 inhibited IL-6-induced TF-1 cell proliferation with an IC ₅₀ value of 7.5 μ M.	
Compound (2a)	2,5-diaminobenzoxazole moiety. Inhibited LPS-induced IL-6 production in Raw264.7 cell line by 29.3 % and IL-6 mRNA inhibition by 44.12 % at 10 μ g/ml.	
Compound (2e)	2,5-diaminobenzoxazole moiety Inhibited LPS-induced IL-6 production in Raw264.7 cell line (10 μ g/ml) and IL-6 mRNA inhibition (10 μ g/ml).	
Compound (3)	Pyrrolidinesulphonylaryl moiety inhibited Hela cell with an EC ₅₀ of 15 μ M.	

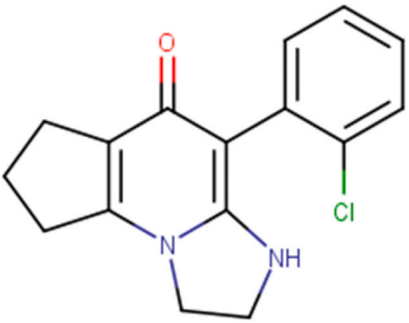
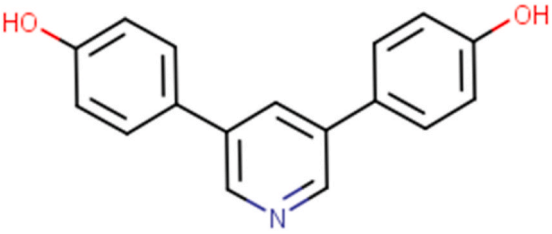
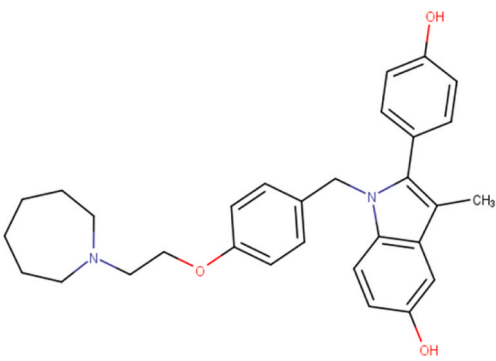
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Table 4 (continued)

Compounds	Cell Lines used for IL-6 inhibition.	Structure
Compound (4)	SC-144 (quinoxalinhydrazide derivative) SC144 Inhibited cell growth in human ovarian cancer cell lines (OVCAR-8, OVCAR-5, OVCAR-3) with IC ₅₀ s (IC ₅₀ = OVCAR-8, OVCAR-5, OVCAR-3 = 0.72, 0.49, 0.95 μ M	
Compound (5)	Raloxifene inhibited L3.6pl or ASPC cells at 10 μ M.	
Compound (6)	DB12903 Developed by pharmacophore modeling with $\Delta G_{\text{bind}} = -118.30$.	
Compound (7)	DB08402 Developed by pharmacophore modeling with $\Delta G_{\text{bind}} = -267.09$.	

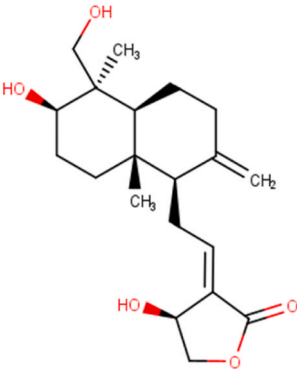
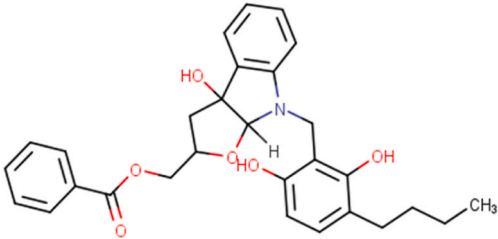
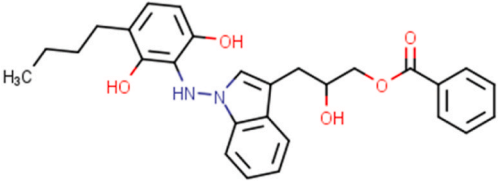
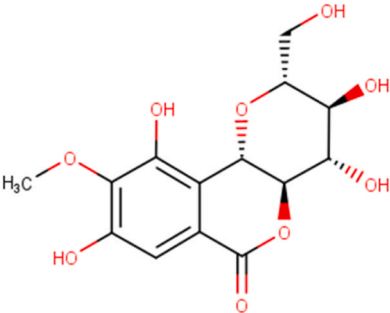
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Table 4 (continued)

Compounds	Cell Lines used for IL-6 inhibition.	Structure
Compound (8) Sch-24471 (2,3,7,8-tetrahydro-4-aryl-1 H-cyclopent[e]imidazo[1,2-a]-pyridin-5(6 H)-one derivatives) decreased IL-6 levels in murine myelomonocytic leukemia (WEH1-265.1) cells by 90 %.		
Compound (9) SCH-21418 (3,5-diarylpyridine derivatives) decreased IL-6 levels in murine myelomonocytic leukemia (WEH1-265.1) cells by 87 %.		
Compound (10) Bazedoxifene inhibited IL-6 in A-549 cell line with an IC50 of 24.7 μM.		

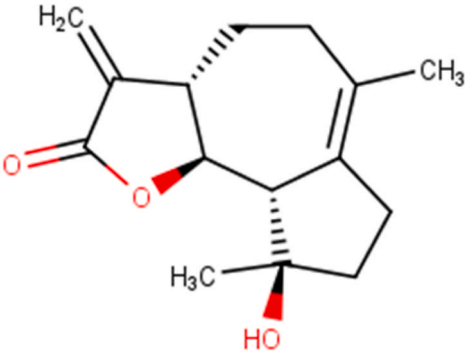
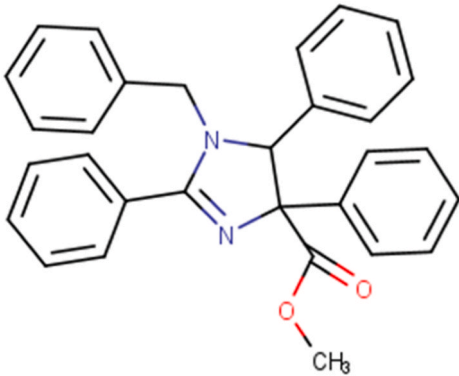
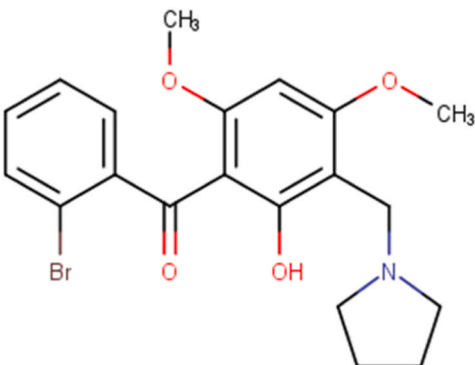
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Table 4 (continued)

Compounds	Cell Lines used for IL-6 inhibition.	Structure
Compound (11)	Andrographolide inhibited LPS-induced IL-6 production in Raw264.7 cell line by 56 %.	
Compound (12)	MDL-5 inhibited IL-6-induced MCF-7 cell proliferation with an IC50 of 5 μ M and MDA-MB-453 cell proliferation with an IC50 of 20 μ M.	
Compound (13)	MDL-16 inhibited IL-6-induced MCF-7 cell proliferation with an IC50 of 20 μ M.	
Compound (14)	Bergenin inhibited IL-6-induced THP-1 cell proliferation with an IC50 of 0.208 μ M.	

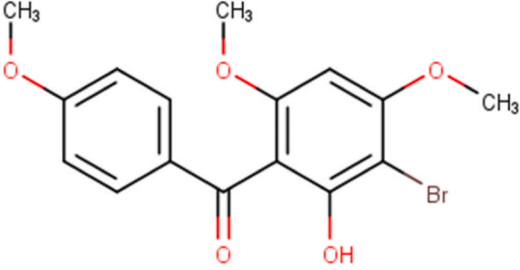
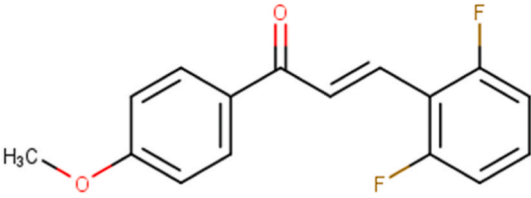
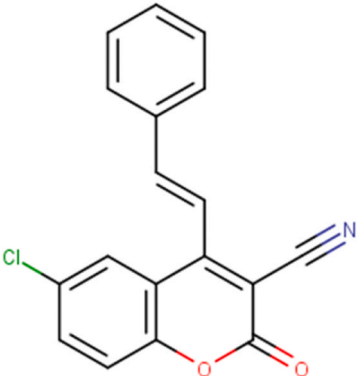
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Table 4 (continued)

Compounds	Cell Lines used for IL-6 inhibition.	Structure
Compound (15)	Micheliolide inhibited gastric cancer in AGS and N87 cell lines at low concentrations.	
Compound (16)	1-benzyl-2,4,5-triphenyl-4,5-dihydro-1 H-imidazole-derived compounds inhibition of NF-κB-mediated IL-6 production which IC50 value is 0.8 μM	
Compound (17)	benzophenone analogues Inhibited IL-6-induced THP-1 cell proliferation by 97 %.	

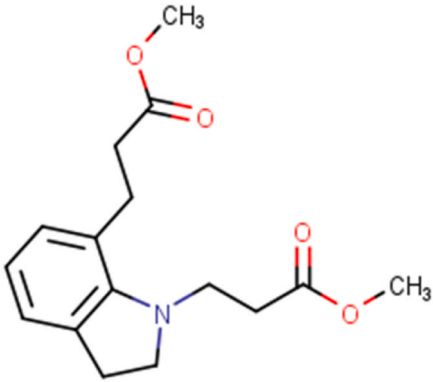
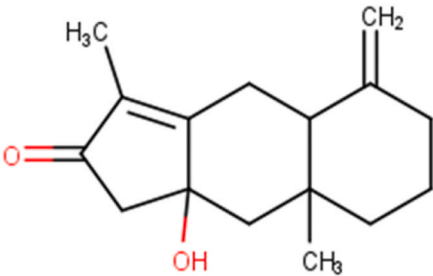
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Table 4 (continued)

Compounds	Cell Lines used for IL-6 inhibition.	Structure
Compound (18)	benzophenone analogues inhibited IL-6-induced THP-1 cell proliferation by 76 %.	
Compound (19)	chalcone-based compounds inhibited LPS-induced production of IL-6 in RAW-264.7 cell line with an IC50 of 5.55 nM.	
Compound (20)	4-styrylcoumarin derivatives inhibited IL-6-induced THP-1 cell proliferation by 87 % at 10 µg/ml.	

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Table 4 (continued)

Compounds	Cell Lines used for IL-6 inhibition.	Structure
Compound (21)	Indoline carbamate derivatives Inhibited LPS-induced production of IL-6 in RAW-264.7 cell line by 45 %.	
Compound (22)	Atractylenolide III Inhibited IL-6 in HMC-1 cell lines.	

signaling axis at a important junction. The molecular specificity of LMT-28, coupled with its demonstrated efficacy in preclinical models, highlights its potential as an innovative therapeutic option for IL-6-driven inflammatory disorders. The compound's capacity to modulate IL-6-induced responses and its promising physicochemical characteristics further support its candidacy as a valuable tool in the development of novel treatments for a range of inflammatory conditions [122]. Yeon-Hwa Park et al. aimed to explore the anti-inflammatory effects of a combination of LMT-28 and metformin in a collagen-induced arthritis (CIA) model, considering the critical role of the IL-6-mediated signaling pathway in rheumatoid arthritis development. The study involved analyzing the impact of the combination treatment on MH7A cell proliferation, IL-6-mediated signaling pathways, and Th17 cell differentiation. Results showed that the combination treatment significantly reduced MH7A cell proliferation, suppressed IL-6-mediated signaling molecules, and attenuated Th17 cell differentiation more effectively than individual treatments. In a CIA mouse model, the combination treatment demonstrated superior efficacy in ameliorating arthritic symptoms, reducing inflammatory cytokine levels, and influencing Treg/Th17 cell distribution. This work suggests that the combined action of LMT-28 and metformin holds promise in mitigating rheumatoid arthritis symptoms by targeting IL-6 signaling and Th17 cell differentiation [123]. Sung-Hoon Ahn et al. conducted a study to characterize the pharmacokinetic properties of LMT-28, a novel synthetic IL-6 inhibitor with potential therapeutic applications in inflammatory diseases. The study encompassed an assessment of LMT-28's physicochemical features, in vitro liver microsomal stability, and in vivo pharmacokinetics using BALB/c mice. LMT-28 exhibited moderate-high cell permeability, as indicated by the partition coefficient (LogP) value of 3.65 and apparent permeability coefficient (Papp) value of 9.7×10^{-6} cm/s. With a 92.4 % plasma protein binding rate, LMT-28 predominantly

bound to serum albumin. The metabolic half-life ($t_{1/2}$) values in rats and humans were 15.3 min and 21.9 min, respectively, at a concentration of 1 μ M. Oral administration (5 mg/kg) of LMT-28 resulted in an area under the plasma drug concentration-time curve of 302 ± 209 ng/ml and a Cmax of 137 ± 100 ng/ml. These findings suggest that LMT-28 possesses favorable physicochemical and pharmacokinetic properties, positioning it as a potential oral drug candidate for synthetic IL-6 inhibition in mammalian inflammation [124]. Qi-qi Liu et al. conducted a study to assess the impact of the GP130-targeting molecule, LMT-28, on lipopolysaccharide (LPS)-induced bone resorption around implants in diabetic models. Osteoblasts stimulated by LPS and advanced glycation end products (AGEs) were treated with LMT-28, revealing downregulation of the nuclear factor- κ B receptor-activating factor ligand (RANKL) expression through the JAK2/STAT3 signaling pathway. In a subsequent in vivo experiment, LMT-28 administration to type 2 diabetic rats with peri-implantitis resulted in reduced bone resorption around implants. Immunohistochemistry showed decreased expression of IL-6 and RANKL, coupled with diminished osteoclast activity. This study underscores the potential of LMT-28 in mitigating LPS-induced bone resorption around implants in diabetic rats [125].

7.2. Diaminobenzoxazole moieties

In the pursuit of developing small molecules targeting inflammatory cytokines for conditions like rheumatoid arthritis (RA), Jihye Yoo et al. synthesized 10 novel compounds, with compound 2e standing out for its potent inhibition (71.5 %) of the Interleukin 6 (IL-6)/Signal Transducer and Activator of Transcription 3 (STAT3) signaling. Moreover, compound 3a demonstrated excellent inhibitory activity (92.1 %) on Interleukin 1 (IL-1 β). Administering compounds 2a and 2e in a zymosan-induced inflammation model in mice significantly reduced

inflammation, particularly with compound 3a exhibiting notable ameliorative effects. These findings suggest that compounds containing the 2,5-diaminobenzoxazole moiety **Table 4, Compound (2a, 2e)** hold promise as small molecule inhibitors for RA treatment, potentially offering advantages over biological drugs [13].

7.3. Pyrrolidinesulphonylaryl molecules

In the pursuit of discovering inhibitors for interleukin-6 (IL-6) signaling, Giovanna Zinzalla et al. synthesized a library of pyrrolidinesulphonylaryl molecules **Table 4, Compound (3)**. Within this library, compound 3 demonstrated notable activity against IL-6/STAT3 signaling at low micromolar levels. Particularly, it selectively inhibited the phosphorylation of STAT3, with a cytostatic effect observed in MDA-MB-231 (STAT3-dependent) breast cancer cells compared to STAT3-null A4 cells. Compound 3 was then investigated in a luciferase reporter assay, revealing its ability to selectively inhibit STAT3 transcriptional activity. Further validation of its specificity was confirmed by its inhibitory effect on STAT3 phosphorylation downstream from IL-6 in both MDA-MB-231 and HeLa cells. Notably, 3 exhibited STAT3-selective properties, distinguishing it from other molecules in the library. This research underscores the potential of compound 3 as a promising candidate for targeted inhibition of IL-6/STAT3 signaling [126].

7.4. Quinoxalinhydrazide derivatives

In this study by Shili Xu et al., the crucial roles of Interleukin (IL)-6 and Stat3 in ovarian cancer progression are highlighted. The signaling axis involving glycoprotein 130 (gp130), a key transducer in this process, is underscored as a potential drug target. Notably, the study introduces SC144, a quinoxalinhydrazide derivative **Table 4, Compound (4)**, as the first-in-class orally active small-molecule inhibitor of gp130. SC144 demonstrates higher potency in human ovarian cancer cell lines compared to normal epithelial cells. Mechanistically, SC144 binds to gp130, leading to phosphorylation (S782) and deglycosylation, thereby disrupting Stat3 phosphorylation and nuclear translocation. Additionally, SC144 inhibits the expression of downstream target genes and effectively hinders gp130 ligand-triggered signaling. Impressively, oral administration of SC144 exhibits significant efficacy in delaying tumor growth in a mouse xenograft model of human ovarian cancer, demonstrating its promising potential as a therapeutic agent with minimal toxicity to normal tissues [127].

7.5. SC144 (quinoxalinhydrazide derivative)

In the study led by Nina A. Hering et al., the focus is on targeting the interleukin-6/glycoprotein-130 signaling pathway to enhance the efficacy of paclitaxel in pancreatic ductal adenocarcinoma (PDAC). Interleukin-6 (IL-6) is identified as a key player in the progression and poor survival of PDAC. The researchers explore the impact of small-molecule inhibitors, specifically the gp130 inhibitor SC144 and raloxifene **Table 4, Compound (5)**, a non-steroidal selective estrogen receptor modulator, on paclitaxel efficacy. The investigation involves in vitro experiments using human pancreatic cell lines (L3.6pl and AsPC-1), employing assays to assess cell viability, proliferation, and apoptosis induction. In vivo studies are conducted using an orthotopic PDAC mouse model, and analyses include qPCR, immunohistochemistry, and ELISA on tumor specimens. The results demonstrate that the combination of paclitaxel with raloxifene, but not SC144, enhances the inhibition of proliferation, viability, and increased apoptosis compared to single treatments in vitro. In the mouse model, both raloxifene/paclitaxel and SC144/paclitaxel combinations reduce tumor weight and volume compared to single-agent therapy or control. Raloxifene/paclitaxel treatment decreases survivin mRNA expression, while SC144/paclitaxel reduces IL-6 levels in tumors and plasma. Overall, the findings suggest

that raloxifene or SC144 can augment the anti-tumorigenic effects of paclitaxel, hinting at the potential to reduce paclitaxel doses in combined chemotherapy to mitigate side effects [128].

7.6. Pharmacophore based molecules

In the work led by queue-Huong Tran et al., the quest for small molecule inhibitors targeting interleukin-6 (IL-6) takes center stage. IL-6, a multifunctional cytokine pivotal for various physiological processes, has gained prominence for its significant role in the cytokine storm observed in COVID-19 patients. While anti-IL-6 receptor monoclonal antibodies show promise in alleviating COVID-19 symptoms, their drawbacks prompt the search for alternative solutions. This study explores potential IL-6 inhibitors by leveraging a novel camelid Fab fragment (68F2) in a crystal protein complex structure (PDB ID: 4ZS7). Pharmacophore models and molecular docking techniques sifted through compounds from DrugBank databases. Oral bioavailability predictions narrowed down the selection to 24 ligands, and subsequent molecular dynamics simulations revealed promising binding affinities. Notably, compounds DB08402 (6) and DB12903 (7) **Table 4, Compound (6,7)**, emerged as frontrunners, demonstrating strong interactions and forming stable complexes with IL-6. These findings lay the groundwork for future developments in small molecule IL-6 inhibitors with the potential to address limitations associated with current treatment approaches [129].

7.7. 2,3,7,8-tetrahydro-4-aryl-1 H-cyclopent[e]imidazo[1,2-a]-pyridin-5(6 H)-one derivatives

Jayaram R. Tagat and colleagues present a study on synthetic inhibitors of Interleukin-6 (IL-6), specifically focusing on 2,3,7,8-tetrahydro-4-aryl-1 H-cyclopent[e]imidazo[1,2-a]-pyridin-5(6 H)-one and related compounds **Table 4, Compound (8)**. The compound Sch-24471 (8) was identified as an IL-6 inhibitor through a screening process involving imidazolines and cyclic beta-keto esters. The researchers systematically explored variations in the core structure I, maintaining its integrity, and studied the impact of changes such as substituents on the phenyl ring, spacers between the tricyclic core and the phenyl ring, and replacements of the phenyl ring with hetero-aromatic rings or other moieties like H, COOR, etc. The resulting series of 2,3,7,8-tetrahydro-4-aryl-1 H-cyclopent[e]imidazo[1,2-a]-pyridin-5(6 H)-one derivatives demonstrated the ability to decrease IL-6 levels in murine myelomonocytic leukemia cells (WEHI-265.1). This study contributes to the development of synthetic IL-6 inhibitors with potential implications for therapeutic interventions targeting IL-6-related pathways [130].

7.8. Diarylpyridine derivatives

In the quest for synthetic inhibitors of interleukin-6 (IL-6), a pivotal cytokine in inflammatory and tumorigenic pathways, SCH-21418 (9) demonstrated significant inhibitory activity. Subsequent systematic synthetic exploration revealed that corresponding meta-terphenyls exhibited comparable efficacy. The synthetic approach involved nickel and palladium-catalyzed coupling of organometallic reagents to aryl halides. The resulting 3,5-dianisylpyridine 5, obtained through demethylation, proved effective with a 70 % yield. Among these compounds, 3,5-diarylpyridine derivatives **Table 4, Compound (9)** demonstrated minimum IL-6 inhibitory activity. Structural modifications, such as the introduction of polar groups or substitution patterns, were explored, highlighting the significance of para-positioned hydroxyl groups on phenyl rings for optimal activity. Analogous compounds exhibited noteworthy in vitro and in vivo IL-6 inhibitory effects. This study unveils promising synthetic IL-6 inhibitors, providing insights for potential therapeutic interventions [131].

7.9. Bazedoxifene-based analogues

Dongmei Song and collaborators report the discovery of bazedoxifene analogues targeting glycoprotein 130 (GP130), a crucial player in signal transduction implicated in various human diseases, particularly cancers. Despite being traditionally considered an undruggable target, the study employs structure-based drug design to create two series of bazedoxifene-based analogues **Table 4, Compound (10)** aiming to disrupt the IL-6/GP130/STAT3 signaling pathway for antitumor treatment. The designed compounds, notably exemplified by compound 10a, demonstrate potent anti-proliferative activity against cancer cells. Compound 10 is shown to directly bind to the GP130 protein, targeting its D1 domain with an affinity (KD) value of 3.8 μ M. Importantly, 10 selectively inhibits JAK2 and STAT3 phosphorylation, as well as IL-6-induced STAT3 phosphorylation, while sparing the D2 and D3 domains of GP130. The compound effectively hinders tumor cell viability, migration, and promotes apoptosis. In vivo, 10 demonstrates efficacy in suppressing xenograft model tumor growth. This study unveils a novel class of bazedoxifene-derived GP130 inhibitors with promising potential as antitumor agents [132].

R. Zhang and colleagues investigated a promising therapeutic approach for ovarian cancer by studying the combined effects of the PARP inhibitor talazoparib and the IL-6 inhibitor baze. Ovarian cancer cells are known for producing IL-6, a factor promoting tumor development and proliferation, as well as contributing to invasive capabilities. Elevated IL-6 levels in ovarian cancer patients are linked to poor prognosis and potential chemoresistance. The study explores the synergistic potential of combining talazoparib, addressing DNA repair mechanisms, with Baze, known for inhibiting IL-6 signaling. This dual-targeting strategy aims to simultaneously impede DNA repair processes and IL-6-mediated tumor growth. Baze, an FDA-approved third-generation selective estrogen receptor modulator, demonstrates a favorable safety profile. The proposed combination therapy suggests a novel approach to ovarian cancer treatment by concurrently tackling DNA repair and IL-6 signaling. However, comprehensive research and clinical trials are essential to validate the efficacy and safety of this therapeutic strategy in ovarian cancer patients [133].

Baze-A, identified in in vitro and in vivo studies, demonstrates the ability to impede the post-attachment step of SARS-CoV-2 entry, highlighting its potential as an antiviral agent with implications for COVID-19 treatment [134]. Clinical investigations revealed a connection between IL-6 and IL-8 levels in ovarian cancer patients' peritoneal fluid and disease progression, indicating their role as prognostic markers. Elevated IL-6 and IL-8 levels are associated with advanced disease stages, higher tumor grades, lymph node metastases, and poor prognosis in ovarian cancer. Notably, Baze-A's inhibitory effects on SARS-CoV-2 and its potential to modulate IL-6 and IL-8 levels suggest broader therapeutic implications beyond its primary indication, offering a promising avenue for further exploration [135]. Inflammation is known to play a critical role in the development of atherosclerosis, particularly in the context of a high-fat diet (HFD) in ApoE/mice. The IL-6/IL-6R/STAT3 signaling pathway has been identified as a key pathway involved in the development and progression of atherosclerosis, making it a potential target for therapeutic interventions. Baze, as an IL-6 inhibitor, has the potential to inhibit the IL-6/IL-6R/STAT3 signaling pathway and thus prevent or attenuate atherosclerosis [136]. Estrogen analogs such as raloxifene and Baze have been developed to address osteoporosis in postmenopausal women. These medications, in addition to their effects on estrogen receptors, have been reported to inhibit IL-6 signaling by interacting with GP130, a component of the IL-6 receptor. IL-6 plays a crucial role in the immune response and has been implicated in the cytokine storm associated with severe COVID-19 cases. Preliminary in-vitro studies have suggested that both Baze and raloxifene possess direct antiviral activity against SARS-CoV-2, the virus causing COVID-19 [137].

7.10. Andrographolide (AP)

Table 4, Compound (11), derived from *Andrographis paniculata*, is a natural compound with a history of traditional use in the preparation Xiyianping (XYP) for treating respiratory infections. XYP, employed for over three decades, harnesses the active ingredients from *Andrographis paniculata*, including 9-dehydro-17-hydro-andrographolide and sodium 9-dehydro-17-hydro-andrographolide-19-yl sulphate. These components are under investigation for their potential anti-inflammatory properties, particularly their ability to inhibit IL-6 production [138, 139]. *Andrographis paniculata*, a medicinal plant, contains a lactone diterpenoid compound called andrographolide [140] which is shown in **Table 4**. In vitro research revealed that andrographolide sulphate demonstrates anti-inflammatory effects by reducing the overproduction of proinflammatory cytokines, including tumor necrosis factor alpha, IL-1, IL-6, and inducible nitric oxide synthase (iNOS). This property led to enhanced survival in mice with LPS-induced sepsis [139]. In a study on *Andrographis paniculata* (AP) derivatives and their impact on LPS-induced IL-6 release in mouse macrophages, ester derivatives, particularly phenyl esters with para-substituents like -NO₂ or -CH₃, demonstrated superior inhibitory effects compared to tricyclic counterparts and alkyl esters. These structural insights highlight the potential of AP derivatives as anti-inflammatory agents targeting IL-6-related disorders [91]. Clinical studies indicate that ParActin® (a standardized *Andrographis paniculata* extract with 50 % andrographolide) provides pain relief in knee osteoarthritis at doses of 300 or 600 mg/day, reducing pain on days 28, 56, and 84 compared to a placebo. ParActin® exhibits anti-inflammatory effects by decreasing markers like COX-2, iNOS, TNF- α , and IL-6 in osteoarthritis, with a favorable safety profile. However, its efficacy in rheumatoid arthritis is less conclusive, with mixed results observed in clinical trials. [141,142].

7.11. Madindoline derivatives

The development of novel small molecules targeting the IL-6/IL-6R/GP130 heterohexamer complex represents a significant breakthrough. Two generations of inhibitors, exemplified by MDL-5 and MDL-16 **Table 4, Compound (12,13)**, have undergone synthesis and evaluation. MDL-5, a first-generation inhibitor, demonstrated reduced STAT3 phosphorylation/activation with IC₅₀ values of 5 μ M and 20 μ M in breast cancer cells (MCF-7 and MDA-MB-453, respectively). MDL-16, a second-generation inhibitor evolved from MDL-5, exhibited improved potency with sub-micro IC₅₀ values, particularly effective in MCF-7 breast cancer cells. Importantly, both compounds showed no induction of cell death in human normal PBMCs. Ongoing tests are expanding to include various breast cancer cell lines, as well as normal cells from the bone marrow and intestinal tract. Furthermore, planned in vivo experiments within the next three months suggest exciting potential for a novel therapeutic option in breast cancer treatment, especially for metastatic and recurrent cases, through targeted IL-6 inhibition [143, 144].

Madindoline A (MDL-A) stands out as a potent and selective inhibitor of interleukin-6 (IL-6) activity, exhibiting superior efficacy compared to its counterpart, madindoline B. With significantly lower IC₅₀ values, approximately five times less than madindoline B, MDL-A demonstrates heightened potency in inhibiting IL-6 function. Notably, MDL-A's specificity extends to gp130 homodimer cytokines, including IL-6 and IL-11, ensuring precise targeting without impacting other signaling pathways or cytokines. Operating through a competitive mechanism, MDL-A competes with IL-6 for receptor binding, effectively thwarting downstream signaling events initiated by IL-6. This targeted approach underscores MDL-A's inhibitory proven against IL-6-mediated processes, marking it as a promising candidate in the realm of IL-6 inhibition. [145,146]. MDL-A, originating from *Streptomyces nitrosporeus* K93-0711, demonstrates notable inhibitory effects on cell lines reliant on IL-6 and IL-11 for growth and survival. This indicates MDL-A's

efficacy in disrupting the signaling pathways governed by these cytokines, resulting in the hindrance of cell proliferation or survival. Intriguingly, the tricyclic moiety of MDL-A alone is insufficient for binding to the target, highlighting the necessity of other structural elements or functional groups within MDL-A for its specific interaction with IL-6 and IL-11 receptors or related components of their signaling pathways. [147]. Indeed, the synthesis of madindoline involves a three-step reaction, proceeding from Nazarov cyclization, Mannich reaction, and deprotection/oxidation, as mentioned in reference [145]. Madindoline A (MDL-A) has demonstrated inhibitory effects on the growth of the HCC cell line HepG2 by targeting IL-6/IL-6R/gp130 trimeric complexes and suppressing their dimerization [146,148]. Elevated levels of IL-6 and IL-11 have been implicated in various diseases, including multiple myeloma, rheumatoid arthritis, hypercalcemia, cancer cachexia, and Castleman's disease. These diseases are associated with dysregulated cytokine signaling and inflammatory responses mediated by IL-6 and IL-11. The development of MDL analogs with improved inhibitory activity against IL-6 and IL-11 holds promise for potential therapeutic applications in these disease conditions [147].

7.12. Bergenin (BG)

Bergenin which is shown in Table 4, Compound (14), is a potent inhibitor of IL-6 with an IC₅₀ value of 0.208 μ M. It is derived from the whole plant ethanolic extract of *Bergenia ciliata* [91]. The rhizome and leaf powder of *Bergenia ciliata* have been used in Ayurveda and traditional medicine for various conditions, including wound healing, earning it the name "Zakhm-e-hayat" (wound healer) in Persian [149]. Bergenin, found in various Indian plant species, exhibits diverse bioavailability. Extracted from *Gendarussa vulgaris* Nees, *Bombax malabaricum* L., *Winter bignonia* (Ker Gawl.) Miers, and more, bergenin's antiparasitic activity varies across sources. It showed limited effectiveness against *P. falciparum* in vitro but inhibited *P. berghei* development in in vivo studies. Extracts from *Rhynchosia aesculifolia* and *Mallotus oppositifolius* leaves demonstrated efficacy against *Plasmodium* and *Leishmania* infections, respectively. Additionally, bergenin-rich extracts exhibited antiviral activity against HSV-1 and influenza-A viruses. In HIV studies, bergenin and norbergenin displayed moderate efficacy. Bergenin also showcased anti-inflammatory effects by suppressing IL-6 and IL-8 production, validated in various in vivo models of inflammation. This multifaceted compound holds promise in parasitic infections, viral diseases, and inflammation, underscoring its pharmacological potential [150]. The anti-arthritis potential of BG has also been recognized. However, its therapeutic effectiveness can be reduced due to decreased absorption and increased instability at neutral-basic pH. In arthritic individuals, higher concentrations of IL-6 have been observed compared to healthy controls. BG and BG-loaded nanoparticles have shown greater ability to block IL-1, IL-6, and TNF- α production compared to standard drugs [151]. Overall, bergenin (BG) shows promise in inhibiting IL-6, exerting antiparasitic effects, and possessing anti-inflammatory and potential anti-arthritis properties.

7.13. Micheliolide derivatives

Micheliolide (MCL) which is shown in Table 4, Compound (15), is a guaianolide sesquiterpene lactone derived from *Michelia compressa* and *Michelia champaca*. It possesses both anti-inflammatory and anti-tumor properties [152]. Elevated levels of the pro-inflammatory cytokine interleukin-6 (IL-6) have been associated with a poor prognosis in gastric cancer patients. MCL has demonstrated anti-cancer effects on stomach cancer by targeting the IL-6/STAT3 pathway, leading to reduced tumor development in both in vitro and in vivo studies [153]. In inflammatory arthritis, such as ankylosing spondylitis (AS), MCL has shown potential therapeutic effects. MCL intervention has been found to decrease the expression levels of caspase-1 p10, IL-1 β p17, NLRP3, caspase-3, and ASC proteins, while inhibiting the levels of TNF- α , IL-1,

IL-6, and IL-18 in AS model mice. This suggests that MCL can alleviate inflammation associated with AS [154]. Diabetic kidney disease (DKD) is a common cause of end-stage renal disease. Dimethylaminomicheliolide, a pro-drug of MCL, has been developed to improve water solubility and achieve sustained release of MCL in the plasma during in vivo administration. Studies have demonstrated that MCL exhibits promising anti-inflammatory effects in cases of DKD. By blocking the expression of metadherin, a protein involved in activating the NF- κ B signaling pathway, MCL inhibits the up-regulation of MCP-1, TNF- α , IL-1 β , and IL-6, which are implicated in the inflammatory processes associated with DKD [155].

7.14. Imidazole-derived compounds

Daljinder K. Kahlon et al. explored a series of 1-benzyl-2,4,5-triphenyl-4,5-dihydro-1 H-imidazole-derived compounds Table 4, Compound (16), incorporating various substituents like ethyl ester, methyl ester, benzyl ester, amide, or alcohol at the 4-position. These compounds demonstrated robust and dose-dependent inhibition of NF- κ B-mediated IL-6 production. The ethyl ester derivative proved to be the most potent, exhibiting an IC₅₀ of 0.8 μ M, and displaying stability against blood esterolytic activity for up to 24 h at 37 °C. Interestingly, compounds with amide and alcohol substitutions showed no activity in inhibiting IL-6 production. Furthermore, the R,R-enantiomer of the ethyl ester derivative demonstrated superior potency (IC₅₀ 0.2 μ M) compared to the S,S-enantiomer (IC₅₀ 0.8 μ M). These findings highlight the potential of these imidazoline scaffolds, particularly the ethyl ester derivative, as promising candidates for inhibiting NF- κ B-mediated IL-6 activity [156].

7.15. Benzophenone analogues

Bandgar et al. conducted a study on the synthesis and biological evaluation of nitrogen-containing benzophenone analogues Table 4, Compound (17) with a focus on their potential as inhibitors for the pro-inflammatory cytokines TNF- α and IL-6, coupled with antioxidant activity. The Mannich reaction was employed for synthesizing these analogues, and their efficacy was assessed in terms of inhibiting TNF- α and IL-6, as well as their antioxidant potential using DPPH radical scavenging assays. The results indicated promising activity against IL-6, ranging from 81 % to 89 % at 10 μ M and 09 % to 42 % at 1 μ M concentration, particularly showcasing inhibitory activity against LPS-induced IL-6 production in the THP-1 cell line. Compound 17 stood out as a particularly effective inhibitor of TNF- α (54 %) and IL-6 (97 %) at 10 μ M, with a still significant inhibition (47 %) at 1 μ M, while exhibiting minimal toxicity (22 %) against CCK-8 cells. Although most compounds demonstrated excellent IL-6 inhibition and moderate to excellent TNF- α inhibition, further optimization is required to address toxicity concerns. Compound 17 displayed notable antioxidant activity among the tested compounds. These findings underscore the potential of certain analogues, particularly 17, as dual inhibitors of TNF- α and IL-6 with antioxidant properties, offering a promising direction for future research and optimization efforts [157].

7.16. Benzophenone derivatives

Bandgar et al. introduced a novel series of benzophenone derivatives Table 4, Compound (18) designed for reduced toxicity and improved pharmacokinetic profiles. The synthesis of these derivatives involved Fridel–Craft acylation, and their evaluation focused on anti-inflammatory activity against Tumor Necrosis Factor- α (TNF- α) and interleukin-6 (IL-6). The LPS-induced cytokine production assay, utilizing phorbol myristate acetate-stimulated human THP-1 cells in vitro, revealed promising IL-6 inhibitory activity among the screened compounds, ranging from 63 % to 82 % at a 10 μ M concentration. Cytotoxicity assessments using CCK-8 cells at 10 μ M indicated that most compounds were not cytotoxic, with benzophenone derivative (18)

standing out for its lack of cytotoxicity up to 100 μM concentration. Notably, compound (18) demonstrated potent IL-6 inhibitory activity, boasting an IC_{50} of 0.19 μM . While generally moderate inhibitors of TNF- α , the synthesized benzophenone derivatives underwent in silico pre-lead prioritization using QikProp 3.2, qualifying them for further investigation. This study presents a noteworthy advancement in the development of benzophenone derivatives with enhanced safety profiles and IL-6 inhibitory potential, paving the way for future exploration in the field [158].

7.17. Chalcone-based compounds

In their study, Wu et al. conducted a screening of a novel series of chalcone-based compounds Table 4, Compound (19) to assess their potential as anti-inflammatory agents targeting LPS-induced IL-6 inhibition in macrophages. The results highlighted the importance of electron-donating substituents in ring A and electron-withdrawing groups in ring B for optimal IL-6 inhibitory effects. The quantitative structure-activity data indicated a preference for substituents on ring A in the order of 3,4,5-(OCH₃)₃ > 3,4-(OCH₃)₂ > 4-OCH₃ > 4-NH₂ and 5-NH₂ > 2-OH > H > 3,5-(F)₂. Additionally, the presence of a halogen on ring B significantly enhanced inhibitory activity. Further mechanistic studies revealed that compound (19) disrupt the JNK/NF- κB signaling mechanism by dose-dependently inhibiting the phosphorylation of JNK and p38. This research provides valuable insights into the structural requirements for effective IL-6 inhibition and elucidates the underlying mechanisms of action for the identified chalcone derivatives, paving the way for the development of novel anti-inflammatory agents [159].

7.18. 4-Styrylcoumarin derivatives

Kuldip Upadhyay and colleagues synthesized a series of 4-styrylcoumarin derivatives Table 4, Compound (20) through Knoevenagel condensation, targeting anti-inflammatory and anti-tubercular activities. The compounds were obtained by reacting substituted 4-methylcoumarin-3-carbonitrile with various heterocyclic or aromatic aldehydes. Evaluation of anti-inflammatory activity against TNF- α and IL-6, as well as anti-tubercular activity, revealed noteworthy findings. Compounds 20 demonstrated significant activity against IL-6 with inhibition of 87 %, while compound 20 exhibited potent activity against TNF- α with a 73 % inhibition at a concentration of 10 μM in THP-1 cells [160].

7.19. Indoline carbamate derivatives

Svetlana Furman and her colleagues explored the synthesis and anti-inflammatory potential of 11 novel indoline carbamate derivatives Table 4, Compound (21). Given the prolonged elevation of pro-inflammatory cytokines, TNF- α , and IL-6 in inflammatory conditions, the team aimed to develop alternatives to existing therapies with potential side effects. The derivatives, incorporating propionic ester, 2-aminoethyl, 3-aminopropyl, 2-(dimethylamino) ethyl, or 3-(dimethylamino)propyl groups at positions 3 or 1, demonstrated notable activities. Specifically, compound (21) previously identified as acetylcholinesterase inhibitors, exhibited concentration-dependent reduction in NO, TNF- α , and IL-6 in LPS-activated RAW-264.7 and mouse peritoneal macrophages at concentrations of 1–10 μM . The anti-inflammatory effects were associated with I $\kappa\text{B}\alpha$ degradation increase and decreased phosphorylation of p38, indicating a potential for these indoline derivatives, particularly those substituted at position 3, in the treatment of chronic inflammatory and neurodegenerative diseases [161].

7.20. Atractylenolide III

In the study conducted by Tae-Hee Kang and colleagues, the sesquiterpenoid Atractylenolide III Table 4, Compound (22), identified as the major bioactive component of *Atractylodes lancea*, exhibited significant regulatory effects on the interleukin (IL)-6 secretion pathway. The research aimed to elucidate the impact of Atractylenolide III on IL-6 secretion and expression in human mast cells (HMC)-1 stimulated with phorbol 12-myristate 13-acetate and calcium ionophore A23187. The sesquiterpenoid demonstrated inhibitory effects on IL-6 secretion and histamine release in stimulated HMC-1 cells. Furthermore, Atractylenolide III suppressed the activation of key signaling molecules, including p38 mitogen-activated protein kinase, C-Jun-N-terminal protein kinase, and nuclear factor- κB , in addition to inhibiting caspase-1 activation and the expression of receptor interacting protein-2. These findings suggest that Atractylenolide III may modulate immunological reactions by regulating IL-6 cellular functions in mast cells, providing valuable insights into its potential therapeutic applications [162].

8. Future aspects

In the future, several key areas of research on IL-6 should be prioritized. Firstly, a deeper understanding of the intricate mechanisms underlying IL-6 signaling and its interactions with other signaling pathways is crucial. By unraveling these complexities, we can gain insights into the precise role of IL-6 in disease pathogenesis and identify potential targets for therapeutic intervention. This knowledge will facilitate the development of more precise and targeted IL-6 inhibitors. Secondly, exploring the potential of combination therapies involving IL-6 inhibitors alongside other therapeutic agents holds great promise. Combinatorial approaches have the potential to synergistically enhance the therapeutic efficacy, overcome resistance mechanisms, and improve patient outcomes. By investigating the interactions between IL-6 inhibitors and other agents, we can uncover novel treatment strategies that maximize the effectiveness of IL-6 blockade. Furthermore, the implementation of personalized medicine approaches in the context of IL-6-targeted therapies is essential. By identifying specific biomarkers or genetic signatures associated with IL-6-mediated diseases, we can better predict individual patient responses and tailor treatment strategies accordingly. This approach has the potential to optimize treatment outcomes, minimize adverse effects, and improve overall patient care. Additionally, comprehensive assessment of the long-term safety and efficacy of IL-6 inhibitors is necessary. Long-term studies and real-world data collection are vital to evaluate the durability of therapeutic responses, potential adverse effects, and the overall impact on patient well-being. This knowledge will guide clinical decision-making and ensure the safe and effective use of IL-6-targeted therapies. Lastly, the discovery and development of novel small molecules with enhanced IL-6 inhibitory properties is a promising avenue for future research. Currently, the market lacks small molecule inhibitors targeting IL-6, and the urgent need for accessible and cost-effective therapies remains. By focusing on the design and optimization of synthetic small molecules, we can expand the repertoire of therapeutic options and provide alternative treatment modalities to complement or even replace current monoclonal antibody-based therapies. In summary, future research on IL-6 should prioritize elucidating signaling mechanisms, exploring combination therapies, implementing personalized medicine approaches, assessing long-term safety and efficacy, and developing novel small molecules. By advancing knowledge in these areas, we can pave the way for more effective and personalized IL-6-targeted therapies, improving patient outcomes in a wide range of IL-6-mediated diseases.

Authors contributions

All authors contributed equally.

Consent for publication

All the authors consent for publication.

Ethics approval and consent to participate.

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Declaration of Competing Interest

The authors declare that there are no conflicts of interest.

Data Availability

Not applicable.

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